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(54) **QUANTUM COMPUTING SYSTEM AND INTER-DEVICE SYNCHRONIZATION CONTROL METHOD FOR QUANTUM COMPUTING**

(71) Applicant: **Hitachi, Ltd.**, Tokyo (JP)

(72) Inventor: **Takanobu TSUNODA**, Tokyo (JP)

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(57) **ABSTRACT**

A quantum computing system includes a quantum device configured to execute a quantum operation; instrumentation devices configured to transmit a control signal for executing the quantum operation; and a synchronization trigger distribution device configured to transmit a second trigger signal for instructing transmission of the control signal to at least one of the instrumentation devices based on a first trigger signal indicating a timing. The synchronization trigger distribution device includes a control sequence table for specifying a predetermined instrumentation device, to which the second trigger signal is to be output, based on the first trigger signal, and a delay time table for specifying a delay time corresponding to each of the instrumentation devices. The synchronization trigger distribution device outputs the second trigger signal to the instrumentation device specified based on the control sequence table, at a timing based on the delay time specified based on the delay time table.

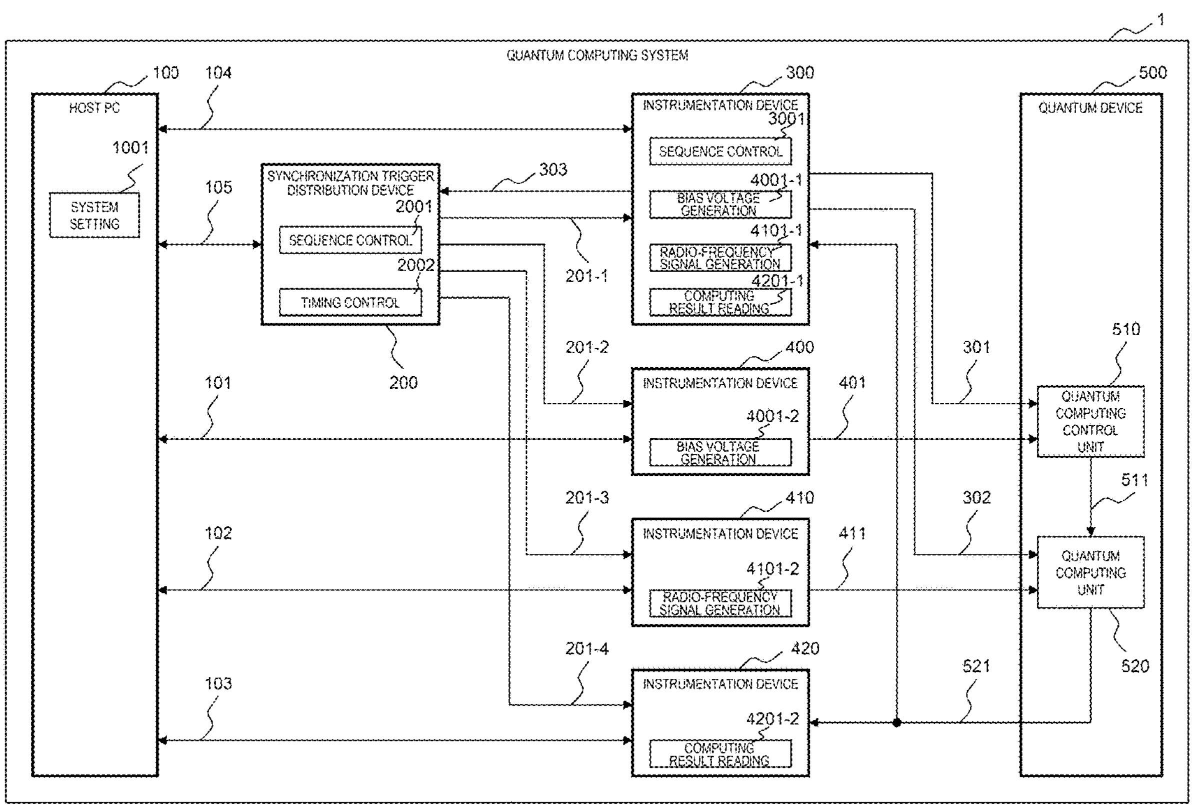


FIG. 1

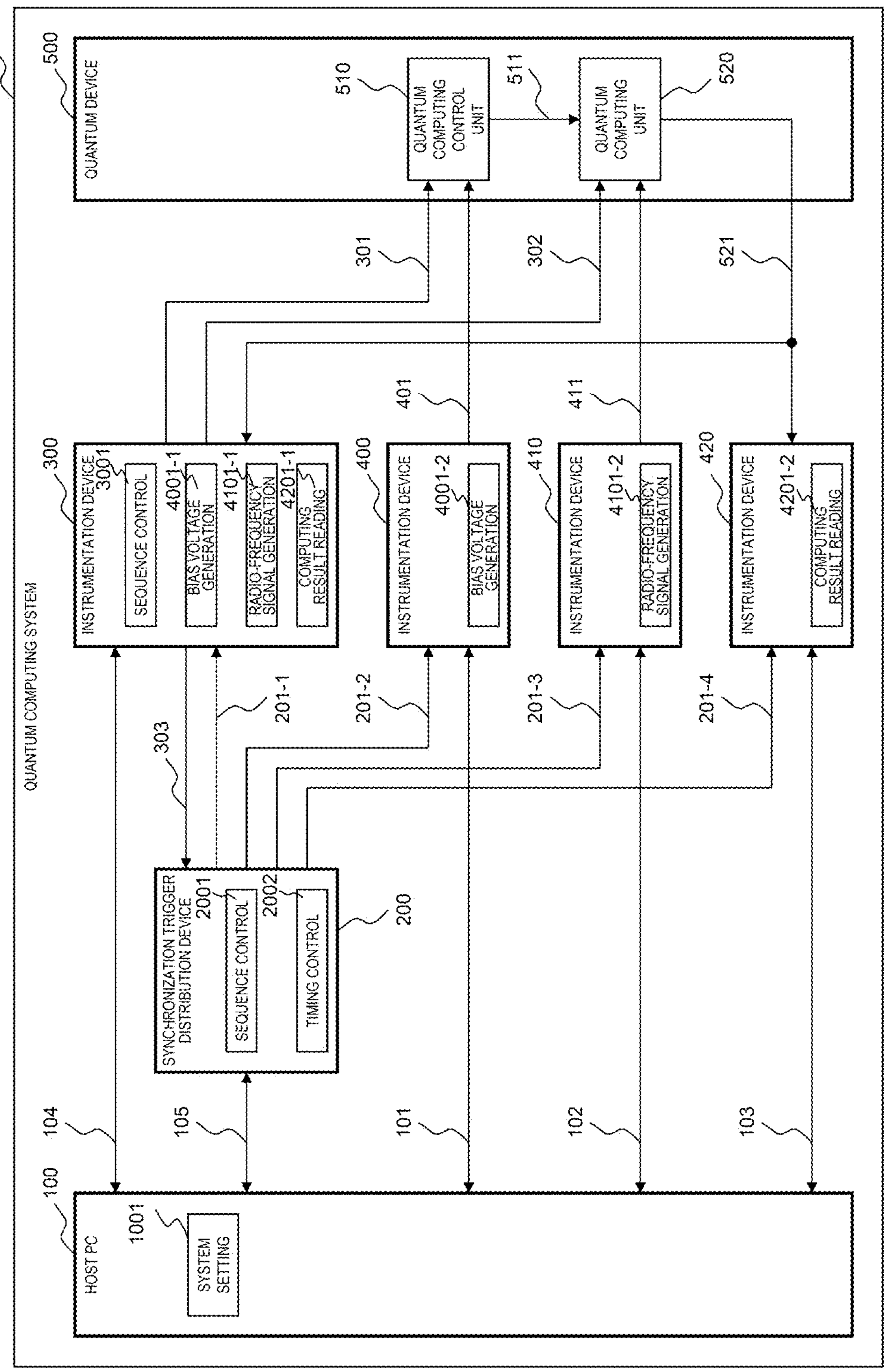

1
QUANTUM COMPUTING SYSTEM
500
QUANTUM DEVICE
510
QUANTUM COMPUTING CONTROL UNIT
511
520
QUANTUM COMPUTING UNIT
301
302
521
300
INSTRUMENTATION DEVICE
3001
SEQUENCE CONTROL
4001-1
BIAS VOLTAGE GENERATION
4101-1
RADIO-FREQUENCY SIGNAL GENERATION
4201-1
COMPUTING RESULT READING
400
INSTRUMENTATION DEVICE
4001-2
BIAS VOLTAGE GENERATION
401
410
INSTRUMENTATION DEVICE
4101-2
RADIO-FREQUENCY SIGNAL GENERATION
411
420
INSTRUMENTATION DEVICE
4201-2
COMPUTING RESULT READING
303
201-1
201-2
201-3
201-4
200
SYNCHRONIZATION TRIGGER DISTRIBUTION DEVICE
2001
SEQUENCE CONTROL
2002
TIMING CONTROL
104
105
101
102
103
100
HOST PC
1001
SYSTEM SETTING

FIG. 2

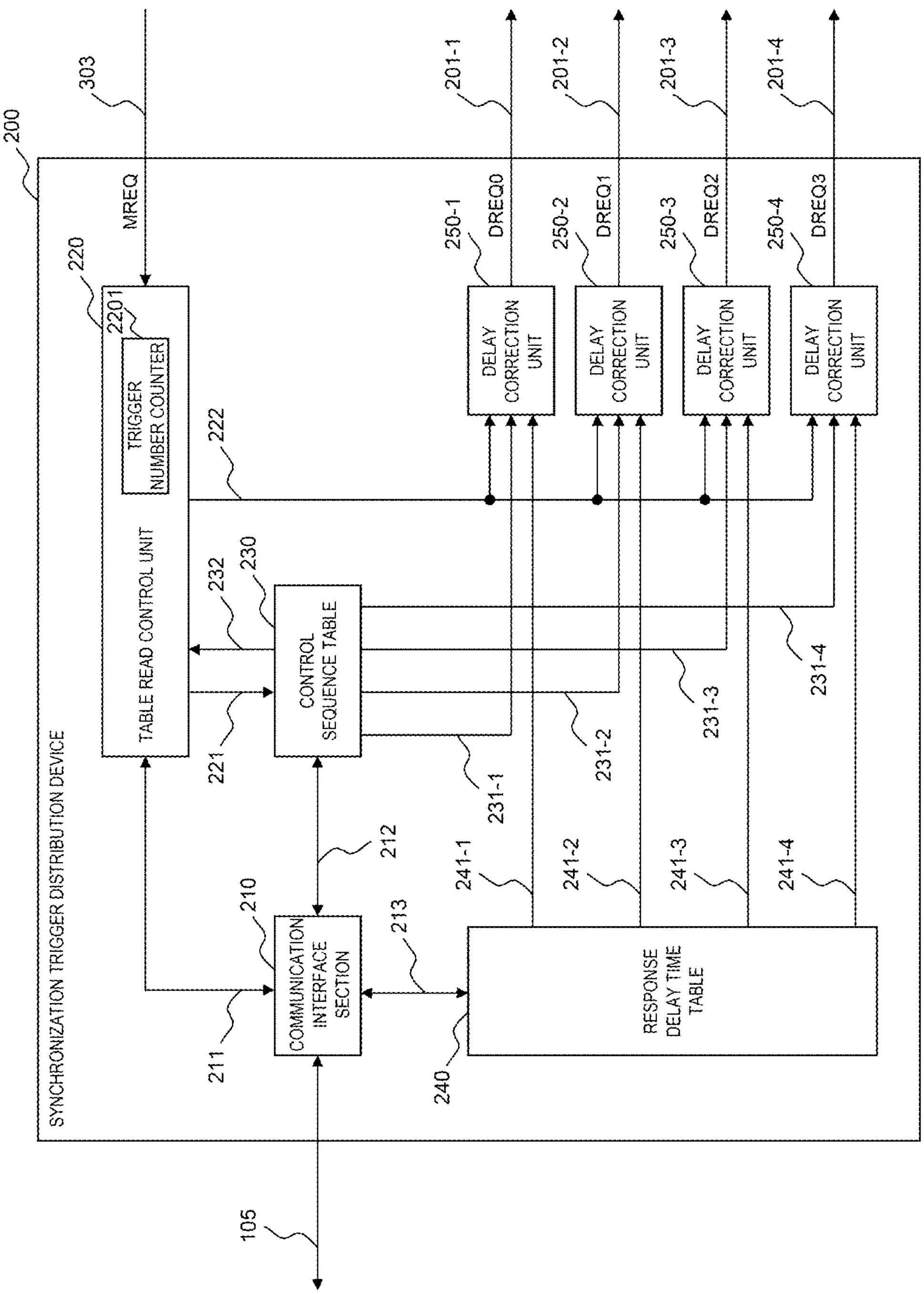

200
SYNCHRONIZATION TRIGGER DISTRIBUTION DEVICE
303
MREQ
220
2201
TRIGGER NUMBER COUNTER
TABLE READ CONTROL UNIT
222
221
232
230
CONTROL SEQUENCE TABLE
231-1
231-2
231-3
231-4
210
COMMUNICATION INTERFACE SECTION
211
212
213
240
RESPONSE DELAY TIME TABLE
241-1
241-2
241-3
241-4
250-1
250-2
250-3
250-4
DELAY CORRECTION UNIT
DELAY CORRECTION UNIT
DELAY CORRECTION UNIT
DELAY CORRECTION UNIT
DREQ0
DREQ1
DREQ2
DREQ3
201-1
201-2
201-3
201-4
105

FIG. 3

230

ENTRY NUMBER (2301)	DISTRIBUTION TRIGGER SIGNAL OUTPUT REQUEST (201-1) (2302-1)	DISTRIBUTION TRIGGER SIGNAL OUTPUT REQUEST (201-2) (2302-2)	DISTRIBUTION TRIGGER SIGNAL OUTPUT REQUEST (201-3) (2302-3)	DISTRIBUTION TRIGGER SIGNAL OUTPUT REQUEST (201-4) (2302-4)	AUXILIARY OPERATION REQUEST (232) (2303)
0	INVALID	INVALID	INVALID	INVALID	ABSENT
1	VALID	INVALID	INVALID	INVALID	ABSENT
2	VALID	VALID	INVALID	INVALID	ABSENT
3	VALID	VALID	VALID	INVALID	ABSENT
4	VALID	INVALID	INVALID	INVALID	ABSENT
5	VALID	INVALID	INVALID	VALID	ABSENT
6	VALID	INVALID	INVALID	INVALID	COMPLETION NOTIFICATION TO HOST PC
⋮	⋮	⋮	⋮	⋮	⋮

FIG. 4

240

2401 2402

TRIGGER SIGNAL TYPE	RESPONSE DELAY TIME
DISTRIBUTION TRIGGER SIGNAL 201-1	160 ns
DISTRIBUTION TRIGGER SIGNAL 201-2	1,000 ns
DISTRIBUTION TRIGGER SIGNAL 201-3	400 ns
DISTRIBUTION TRIGGER SIGNAL 201-4	640 ns

FIG. 5

COMMAND	FORMAT	OPERATION DEFINITION
SETCNTR	11110_mmmmmm(c1)_nnnnnn(c2)_1	SET TRIGGER NUMBER COUNT VALUE OF TABLE READ CONTROL UNIT 220 (OPTION) mmmmmm: UPPER 6 BITS OF COUNTER VALUE c1: CHECK BIT FOR UPPER 6 BITS nnnnnn: LOWER 6 BITS OF COUNTER VALUE c2: CHECK BIT FOR LOWER 6 BITS
CHKCNTR	11111_mmmmmm(c1)_nnnnnn(c2)_1	NOTIFY EXPECTED VALUE OF TRIGGER NUMBER COUNT VALUE OF TABLE READ CONTROL UNIT 220 (OPTION) mmmmmm: UPPER 6 BITS OF COUNTER VALUE c1: CHECK BIT FOR UPPER 6 BITS nnnnnn: LOWER 6 BITS OF COUNTER VALUE c2: CHECK BIT FOR LOWER 6 BITS
NEXT	110	INCREMENT TRIGGER NUMBER COUNTER SET OPERATING STATE OF TABLE READ CONTROL UNIT 220 TO SEQUENCE EXECUTION STATE
END	1110	INCREMENT TRIGGER NUMBER COUNTER SET OPERATING STATE OF TABLE READ CONTROL UNIT 220 TO SEQUENCE NON-EXECUTION STATE

FIG. 6

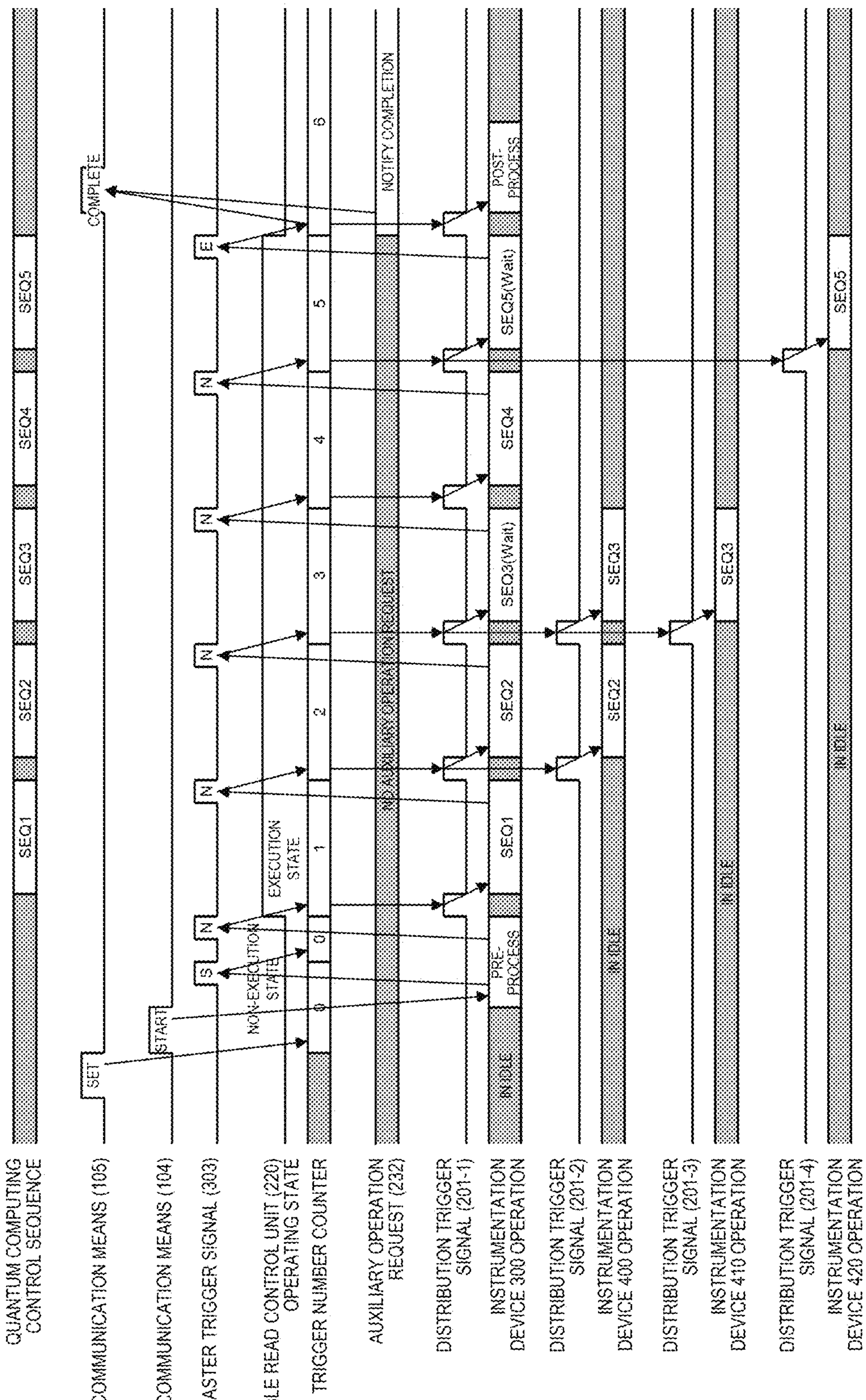

QUANTUM COMPUTING CONTROL SEQUENCE
COMMUNICATION MEANS (105)
COMMUNICATION MEANS (104)
MASTER TRIGGER SIGNAL (303)
TABLE READ CONTROL UNIT (220) OPERATING STATE
TRIGGER NUMBER COUNTER
AUXILIARY OPERATION REQUEST (232)
DISTRIBUTION TRIGGER SIGNAL (201-1)
INSTRUMENTATION DEVICE 300 OPERATION
DISTRIBUTION TRIGGER SIGNAL (201-2)
INSTRUMENTATION DEVICE 400 OPERATION
DISTRIBUTION TRIGGER SIGNAL (201-3)
INSTRUMENTATION DEVICE 410 OPERATION
DISTRIBUTION TRIGGER SIGNAL (201-4)
INSTRUMENTATION DEVICE 420 OPERATION
SET
START
COMPLETE
S
N
E
NON-EXECUTION STATE
EXECUTION STATE
0
1
2
3
4
5
6
SEQ1
SEQ2
SEQ3
SEQ4
SEQ5
NO AUXILIARY OPERATION REQUEST
NOTIFY COMPLETION
IN IDLE
PRE-PROCESS
SEQ3(Wait)
SEQ5(Wait)
POST-PROCESS

FIG. 7

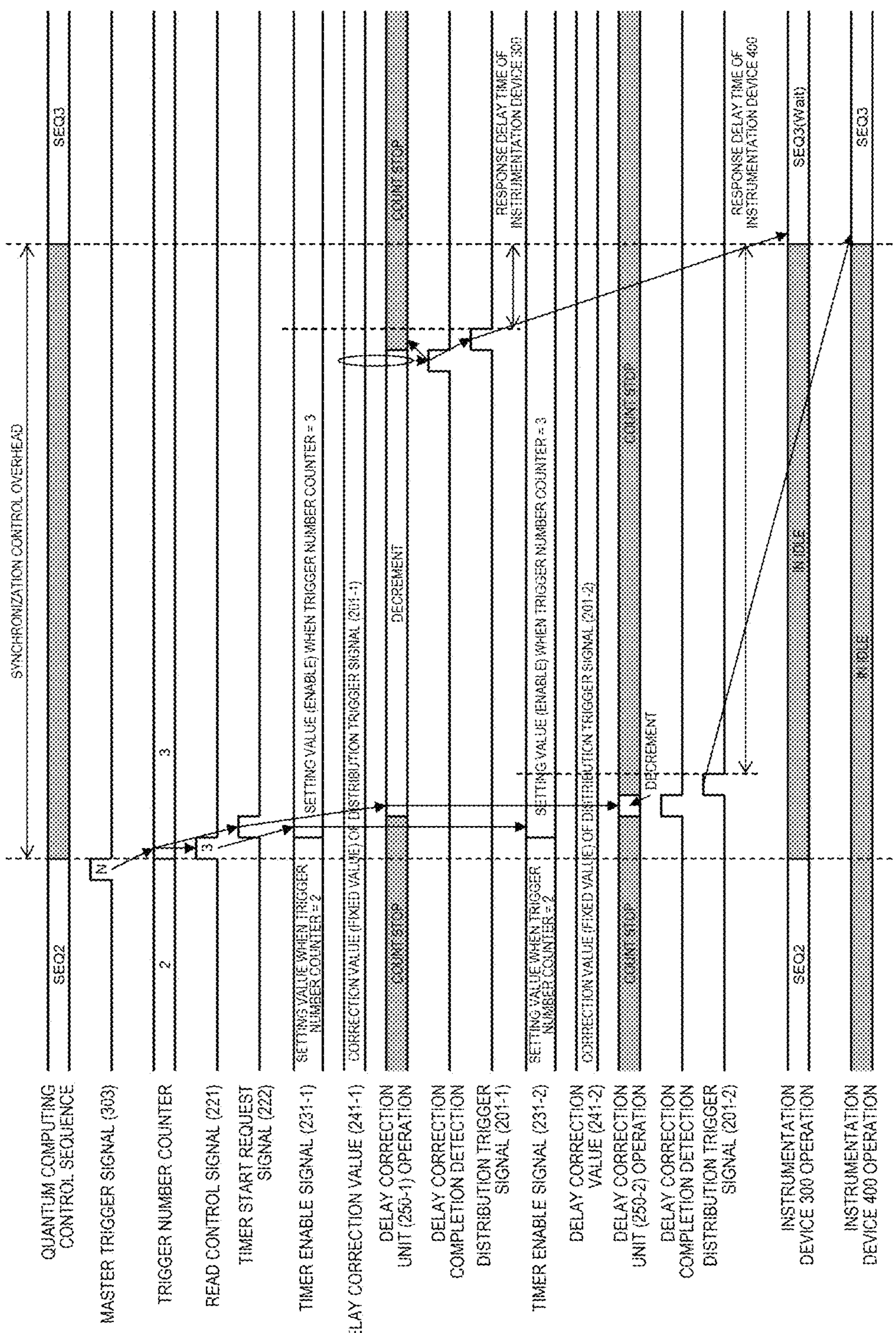

QUANTUM COMPUTING CONTROL SEQUENCE
MASTER TRIGGER SIGNAL (303)
TRIGGER NUMBER COUNTER
READ CONTROL SIGNAL (221)
TIMER START REQUEST SIGNAL (222)
TIMER ENABLE SIGNAL (231-1)
DELAY CORRECTION VALUE (241-1)
DELAY CORRECTION UNIT (250-1) OPERATION
DELAY CORRECTION COMPLETION DETECTION
DISTRIBUTION TRIGGER SIGNAL (201-1)
TIMER ENABLE SIGNAL (231-2)
DELAY CORRECTION VALUE (241-2)
DELAY CORRECTION UNIT (250-2) OPERATION
DELAY CORRECTION COMPLETION DETECTION
DISTRIBUTION TRIGGER SIGNAL (201-2)
INSTRUMENTATION DEVICE 300 OPERATION
INSTRUMENTATION DEVICE 400 OPERATION
SEQ2
SEQ3
SYNCHRONIZATION CONTROL OVERHEAD
N
2
3
SETTING VALUE WHEN TRIGGER NUMBER COUNTER = 2
SETTING VALUE (ENABLE) WHEN TRIGGER NUMBER COUNTER = 3
CORRECTION VALUE (FIXED VALUE) OF DISTRIBUTION TRIGGER SIGNAL (201-1)
CORRECTION VALUE (FIXED VALUE) OF DISTRIBUTION TRIGGER SIGNAL (201-2)
COUNT STOP
DECREMENT
RESPONSE DELAY TIME OF INSTRUMENTATION DEVICE 300
RESPONSE DELAY TIME OF INSTRUMENTATION DEVICE 400
SEQ3(Wait)
IDLE

FIG. 8

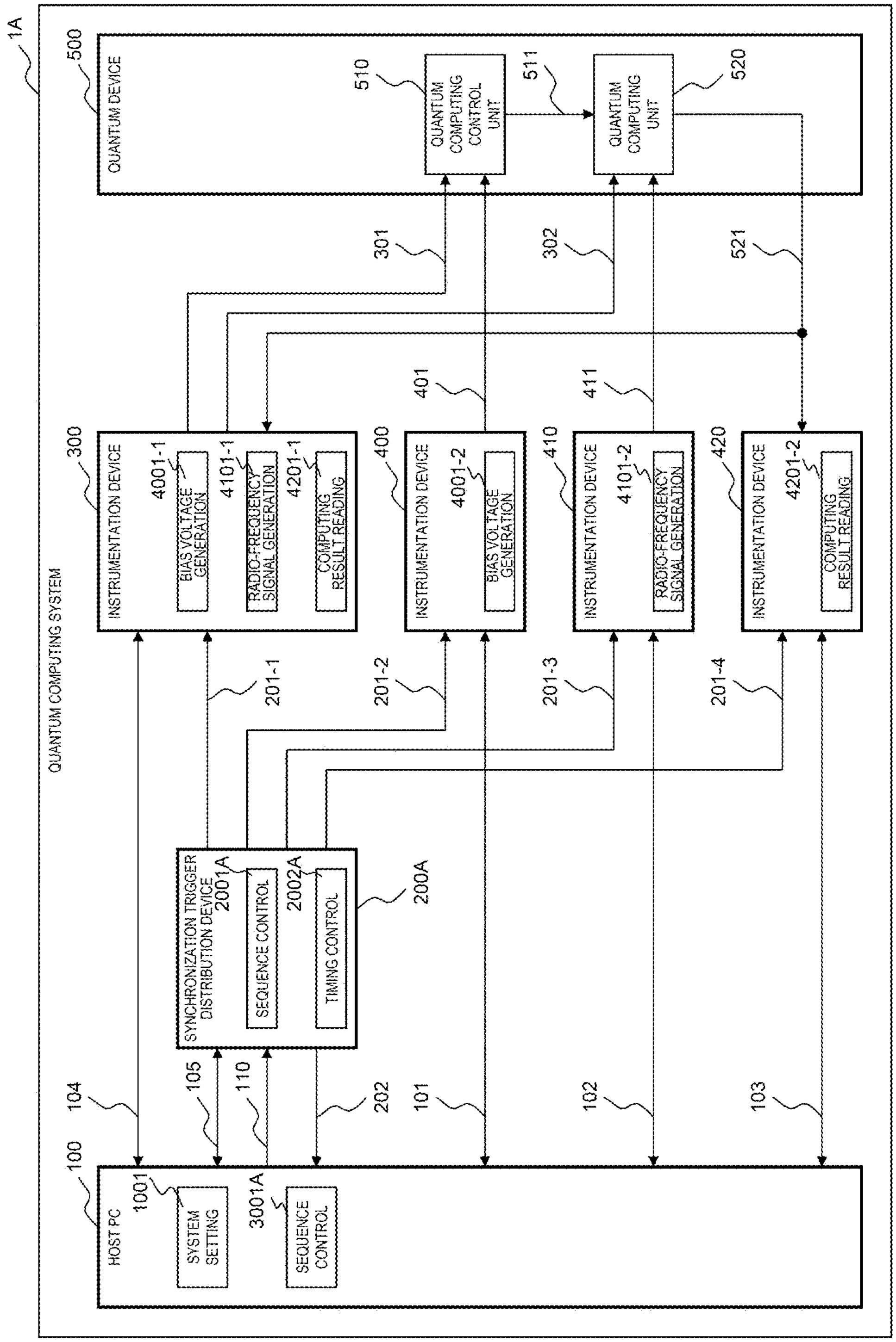

1A
QUANTUM COMPUTING SYSTEM
500
QUANTUM DEVICE
510
QUANTUM COMPUTING CONTROL UNIT
511
520
QUANTUM COMPUTING UNIT
301
302
521
401
411
300
INSTRUMENTATION DEVICE
4001-1
BIAS VOLTAGE GENERATION
4101-1
RADIO-FREQUENCY SIGNAL GENERATION
4201-1
COMPUTING RESULT READING
400
INSTRUMENTATION DEVICE
4001-2
BIAS VOLTAGE GENERATION
410
INSTRUMENTATION DEVICE
4101-2
RADIO-FREQUENCY SIGNAL GENERATION
420
INSTRUMENTATION DEVICE
4201-2
COMPUTING RESULT READING
201-1
201-2
201-3
201-4
SYNCHRONIZATION TRIGGER DISTRIBUTION DEVICE
2001A
SEQUENCE CONTROL
2002A
TIMING CONTROL
200A
104
105
110
202
101
102
103
100
HOST PC
1001
SYSTEM SETTING
3001A
SEQUENCE CONTROL

FIG. 9

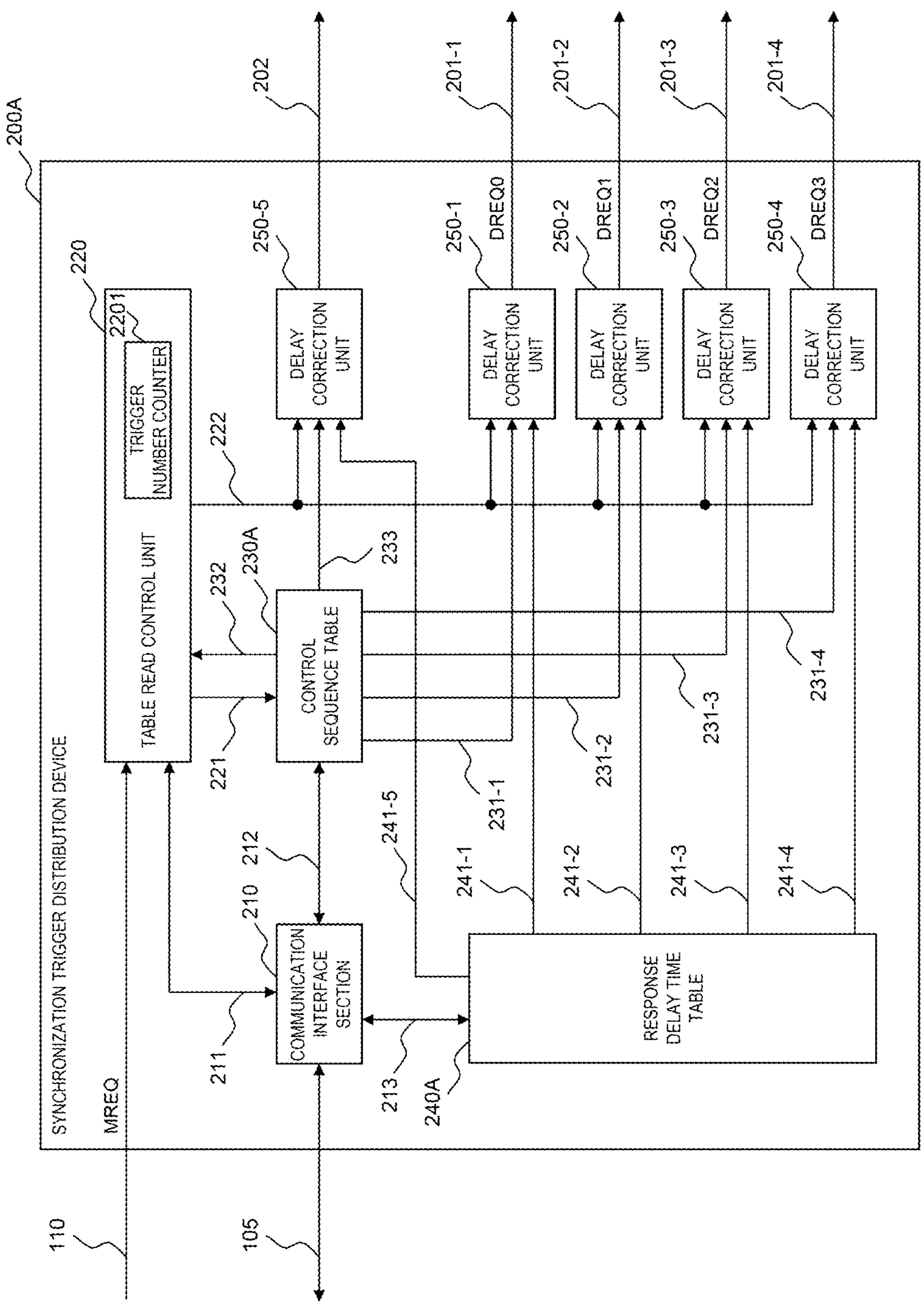

200A
SYNCHRONIZATION TRIGGER DISTRIBUTION DEVICE
MREQ
110
105
211
210
COMMUNICATION INTERFACE SECTION
212
213
240A
RESPONSE DELAY TIME TABLE
241-1
241-2
241-3
241-4
241-5
221
TABLE READ CONTROL UNIT
232
230A
CONTROL SEQUENCE TABLE
233
231-1
231-2
231-3
231-4
220
2201
TRIGGER NUMBER COUNTER
222
DELAY CORRECTION UNIT
250-5
250-1
250-2
250-3
250-4
DREQ0
DREQ1
DREQ2
DREQ3
202
201-1
201-2
201-3
201-4

FIG. 10

230A

2301 ENTRY NUMBER	2302-1 DISTRIBUTION TRIGGER SIGNAL OUTPUT REQUEST (201-1)	2302-2 DISTRIBUTION TRIGGER SIGNAL OUTPUT REQUEST (201-2)	2302-3 DISTRIBUTION TRIGGER SIGNAL OUTPUT REQUEST (201-3)	2302-4 DISTRIBUTION TRIGGER SIGNAL OUTPUT REQUEST (201-4)	2302-5 DISTRIBUTION TRIGGER SIGNAL OUTPUT REQUEST (202)	2303 AUXILIARY OPERATION REQUEST (232)
0	INVALID	INVALID	INVALID	INVALID	INVALID	ABSENT
1	VALID	INVALID	INVALID	INVALID	VALID	ABSENT
2	VALID	VALID	INVALID	INVALID	VALID	ABSENT
3	INVALID	VALID	VALID	INVALID	VALID	ABSENT
4	VALID	INVALID	INVALID	INVALID	VALID	ABSENT
5	INVALID	INVALID	INVALID	VALID	VALID	ABSENT
6	INVALID	INVALID	INVALID	INVALID	VALID	ABSENT
⋮	⋮	⋮	⋮	⋮	⋮	⋮

FIG. 11

240A

2401 2402

TRIGGER SIGNAL TYPE	RESPONSE DELAY TIME
DISTRIBUTION TRIGGER SIGNAL 201-1	160 ns
DISTRIBUTION TRIGGER SIGNAL 201-2	1,000 ns
DISTRIBUTION TRIGGER SIGNAL 201-3	400 ns
DISTRIBUTION TRIGGER SIGNAL 201-4	640 ns
DISTRIBUTION TRIGGER SIGNAL 202	20 ns

FIG. 12

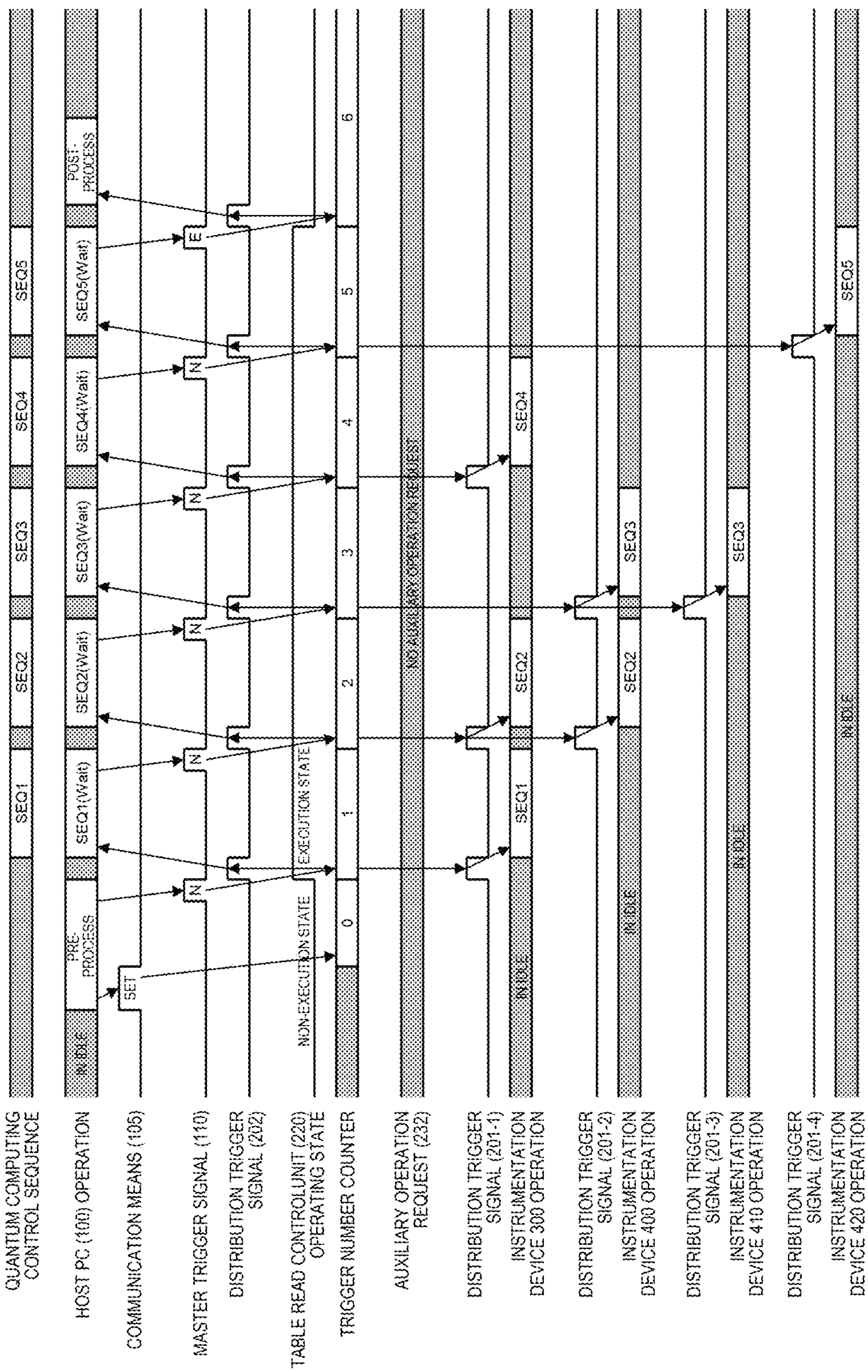

QUANTUM COMPUTING CONTROL SEQUENCE
HOST PC (100) OPERATION
COMMUNICATION MEANS (105)
MASTER TRIGGER SIGNAL (110)
DISTRIBUTION TRIGGER SIGNAL (202)
TABLE READ CONTROLUNIT (220) OPERATING STATE
TRIGGER NUMBER COUNTER
AUXILIARY OPERATION REQUEST (232)
DISTRIBUTION TRIGGER SIGNAL (201-1)
INSTRUMENTATION DEVICE 300 OPERATION
DISTRIBUTION TRIGGER SIGNAL (201-2)
INSTRUMENTATION DEVICE 400 OPERATION
DISTRIBUTION TRIGGER SIGNAL (201-3)
INSTRUMENTATION DEVICE 410 OPERATION
DISTRIBUTION TRIGGER SIGNAL (201-4)
INSTRUMENTATION DEVICE 420 OPERATION
SEQ1
SEQ2
SEQ3
SEQ4
SEQ5
IN IDLE
PRE-PROCESS
SEQ1(Wait)
SEQ2(Wait)
SEQ3(Wait)
SEQ4(Wait)
SEQ5(Wait)
POST-PROCESS
SET
N
E
NON-EXECUTION STATE
EXECUTION STATE
0
1
2
3
4
5
6
NO AUXILIARY OPERATION REQUEST

FIG. 13

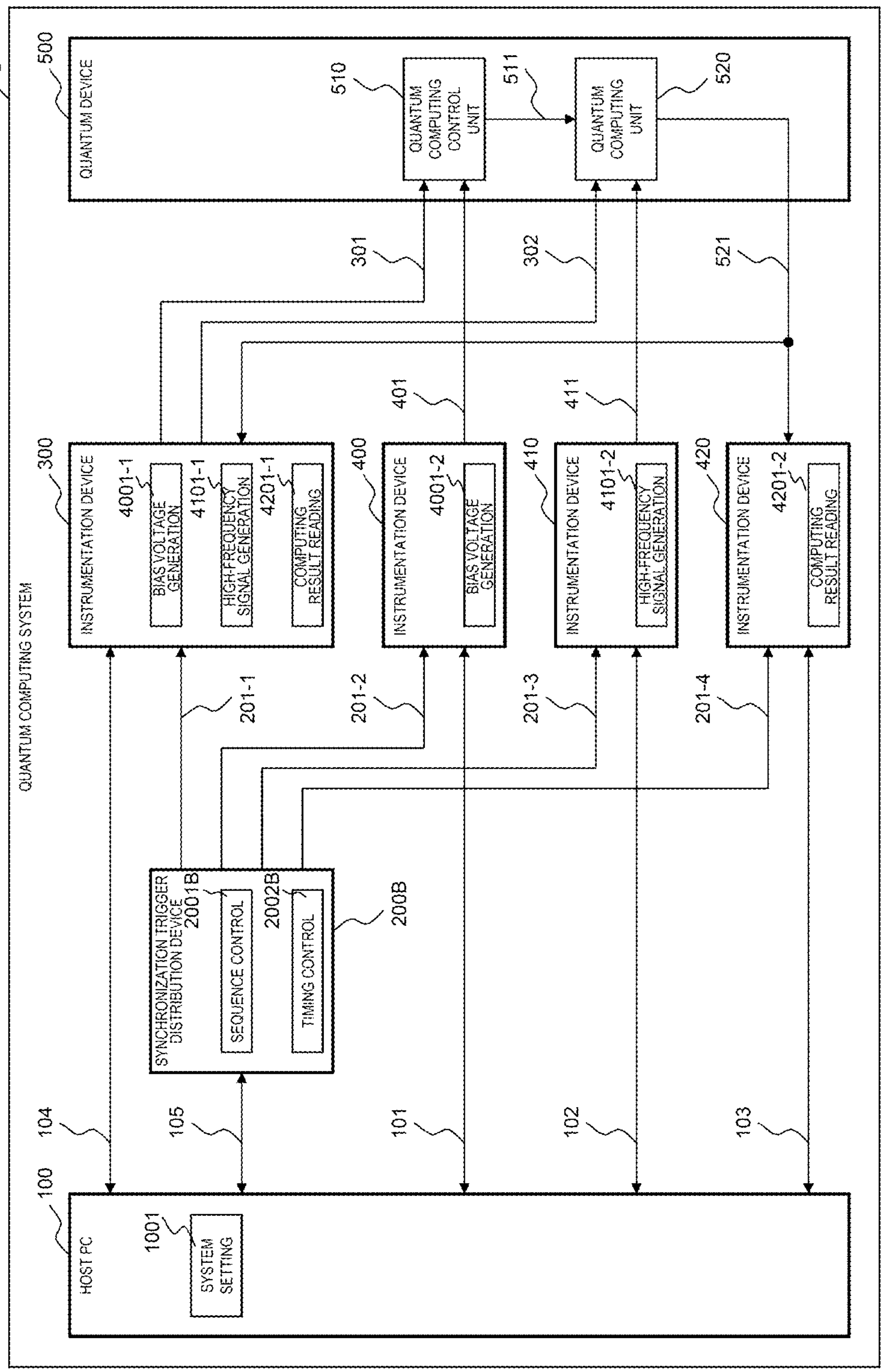

1B
QUANTUM COMPUTING SYSTEM
500
QUANTUM DEVICE
510
QUANTUM COMPUTING CONTROL UNIT
511
520
QUANTUM COMPUTING UNIT
301
302
521
300
INSTRUMENTATION DEVICE
4001-1
BIAS VOLTAGE GENERATION
4101-1
HIGH-FREQUENCY SIGNAL GENERATION
4201-1
COMPUTING RESULT READING
400
INSTRUMENTATION DEVICE
4001-2
BIAS VOLTAGE GENERATION
401
410
INSTRUMENTATION DEVICE
4101-2
HIGH-FREQUENCY SIGNAL GENERATION
411
420
INSTRUMENTATION DEVICE
4201-2
COMPUTING RESULT READING
201-1
201-2
201-3
201-4
SYNCHRONIZATION TRIGGER DISTRIBUTION DEVICE
2001B
SEQUENCE CONTROL
2002B
TIMING CONTROL
200B
104
105
101
102
103
100
HOST PC
1001
SYSTEM SETTING

FIG. 14

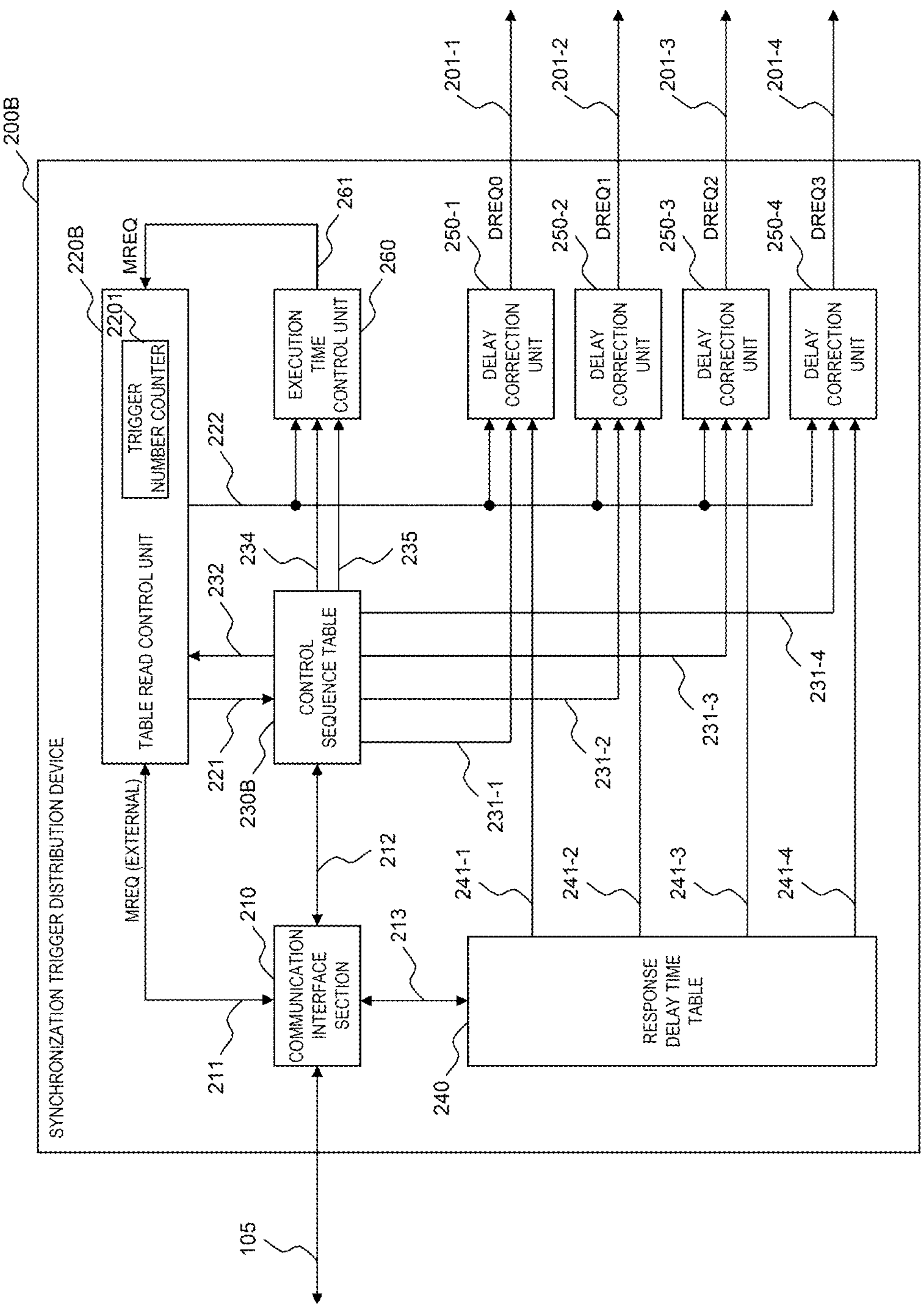

200B
SYNCHRONIZATION TRIGGER DISTRIBUTION DEVICE
220B
2201
MREQ
TRIGGER NUMBER COUNTER
TABLE READ CONTROL UNIT
EXECUTION TIME CONTROL UNIT
261
260
DELAY CORRECTION UNIT
250-1
250-2
250-3
250-4
DREQ0
DREQ1
DREQ2
DREQ3
201-1
201-2
201-3
201-4
222
232
234
235
CONTROL SEQUENCE TABLE
221
230B
231-1
231-2
231-3
231-4
MREQ (EXTERNAL)
210
212
COMMUNICATION INTERFACE SECTION
211
213
RESPONSE DELAY TIME TABLE
240
241-1
241-2
241-3
241-4
105

FIG. 15

230B

ENTRY NUMBER	DISTRIBUTION TRIGGER SIGNAL OUTPUT REQUEST (201-1) [2301]	DISTRIBUTION TRIGGER SIGNAL OUTPUT REQUEST (201-2) [2302-1]	DISTRIBUTION TRIGGER SIGNAL OUTPUT REQUEST (201-3) [2302-2]	DISTRIBUTION TRIGGER SIGNAL OUTPUT REQUEST (201-4) [2302-3]	MASTER TRIGGER SIGNAL OUTPUT REQUEST (261) [2302-4]	EXECUTION TIME SETTING VALUE (234) [2304]	AUXILIARY OPERATION REQUEST (232) [2305]
0	INVALID	INVALID	INVALID	INVALID	VALID	4,000 ns	ABSENT
1	VALID	INVALID	INVALID	INVALID	VALID	2,000 ns	ABSENT
2	VALID	VALID	INVALID	INVALID	VALID	3,000 ns	ABSENT
3	INVALID	VALID	VALID	INVALID	VALID	1,200 ns	ABSENT
4	VALID	INVALID	INVALID	INVALID	VALID	2,500 ns	ABSENT
5	INVALID	INVALID	INVALID	VALID	VALID	100 ns	ABSENT
6	INVALID	INVALID	INVALID	INVALID	INVALID	0 ns	COMPLETION NOTIFICATION TO HOST PC SHIFT TO NON-EXECUTION STATE
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

2303

FIG. 16

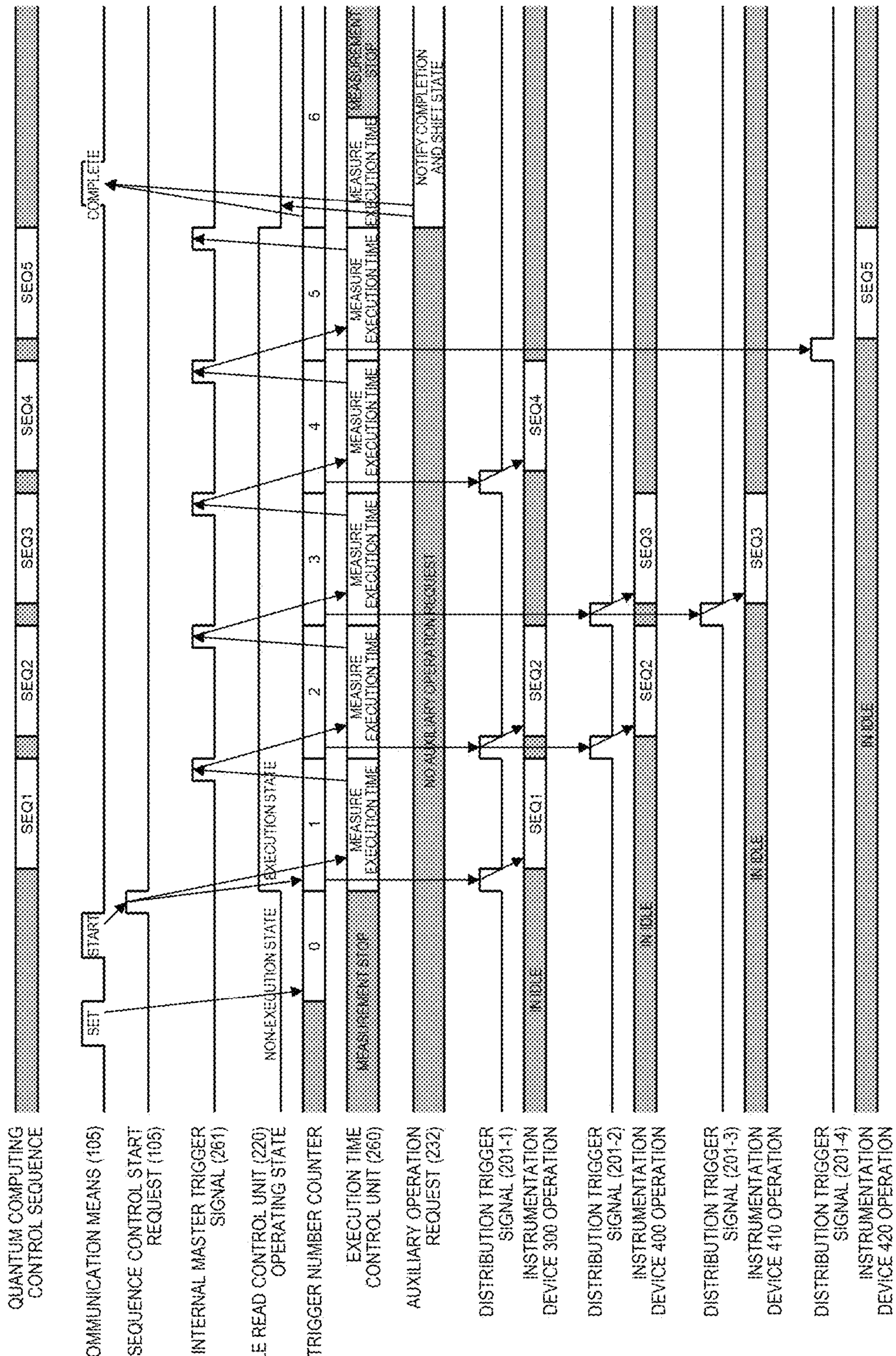
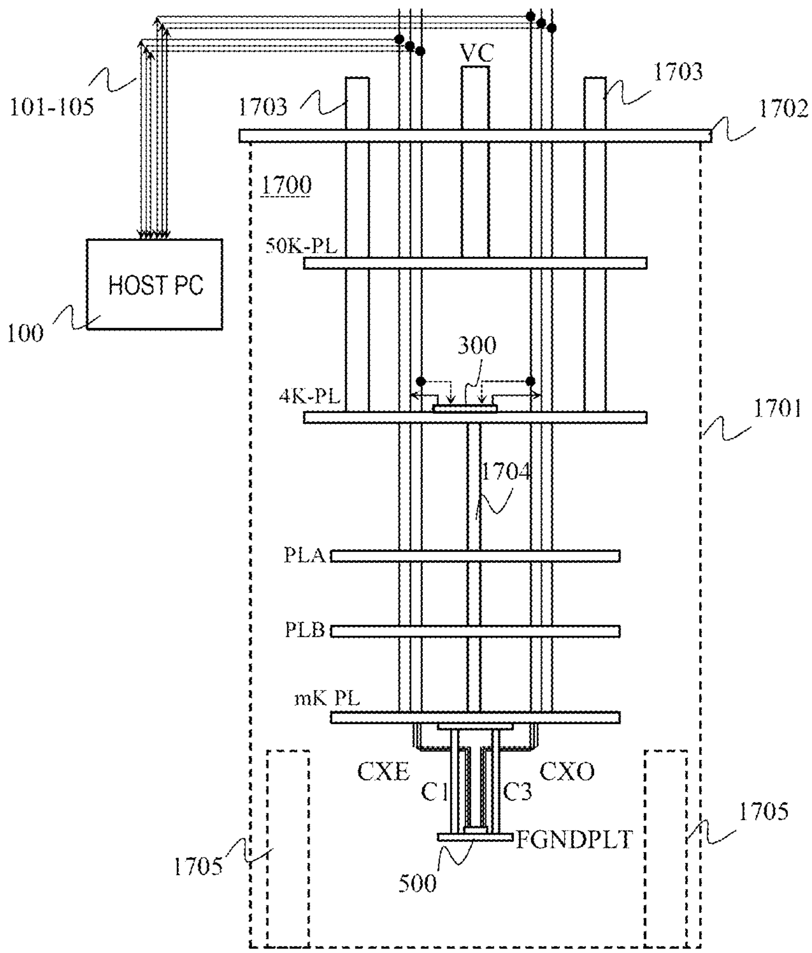

QUANTUM COMPUTING CONTROL SEQUENCE
COMMUNICATION MEANS (105)
SEQUENCE CONTROL START REQUEST (105)
INTERNAL MASTER TRIGGER SIGNAL (261)
TABLE READ CONTROL UNIT (220) OPERATING STATE
TRIGGER NUMBER COUNTER
EXECUTION TIME CONTROL UNIT (260)
AUXILIARY OPERATION REQUEST (232)
DISTRIBUTION TRIGGER SIGNAL (201-1)
INSTRUMENTATION DEVICE 300 OPERATION
DISTRIBUTION TRIGGER SIGNAL (201-2)
INSTRUMENTATION DEVICE 400 OPERATION
DISTRIBUTION TRIGGER SIGNAL (201-3)
INSTRUMENTATION DEVICE 410 OPERATION
DISTRIBUTION TRIGGER SIGNAL (201-4)
INSTRUMENTATION DEVICE 420 OPERATION
SET
START
COMPLETE
SEQ1
SEQ2
SEQ3
SEQ4
SEQ5
NON-EXECUTION STATE
EXECUTION STATE
0
1
2
3
4
5
6
MEASUREMENT STOP
MEASURE EXECUTION TIME
NO AUXILIARY OPERATION REQUEST
NOTIFY COMPLETION AND SHIFT STATE
IN IDLE

FIG. 17


VC
1703
1703
101-105
1702
1700
50K-PL
HOST PC
100
300
4K-PL
1701
1704
PLA
PLB
mK PL
CXE
CXO
C1
C3
1705
1705
FGNDPLT
500

QUANTUM COMPUTING SYSTEM AND INTER-DEVICE SYNCHRONIZATION CONTROL METHOD FOR QUANTUM COMPUTING

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority from Japanese application JP2023-171298, filed on Oct. 2, 2023, the content of which is hereby incorporated by reference into this application.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates to a gate type quantum computer system, and more particularly, to an inter-device synchronization control method for constructing a system by a plurality of instrumentation devices (including a form of a semiconductor chip), which operate in cooperation with each other with high timing accuracy, by using an operation for an electron spin of a quantum computing device in which a plurality of single electron spin quantum dots are arranged and generating a control signal necessary for reading an electron spin state.

2. Description of Related Art

[0003] It is seen as a sure thing that the performance improvement by the miniaturization of the semiconductor element which has supported the progress of the computer for more than a half century will reach its limit in the near future considering the comparison between the process rules and the interatomic distance of silicon. Quantum computers are an attempt to break the limit by new computing principles and devices, and quantum computing devices and computing methods using superconducting circuits, ion traps, photons, silicon quantum dots, and the like have been proposed. In addition, as a first step to a future concept of executing a large-scale practical application in a general-purpose quantum computer system having error tolerance, the principle proof and algorithm search for a system referred to as a noisy intermediate-scale quantum device (NISQ) are in progress on the premise of a concept at the present time that the number of quantum bits is as small as about 100 and error correction cannot be performed.

[0004] In order to implement a quantum computer system to which a quantum computing device using a quantum effect exhibited at an extremely low temperature as a computing principle, such as a single electron spin in a silicon quantum dot, a dilution refrigerator is generally selected as an accommodating device. In this case, a configuration of an experimental device is often adopted in which the quantum computing device is fixed in thermal contact with a mixing chamber in which the temperature is lowest in the dilution refrigerator, a signal necessary for operation control of the quantum computing device is applied from a signal generator provided outside the dilution refrigerator, and a computing result (quantum dot state) is read by a measuring machine provided similarly outside the dilution refrigerator.

[0005] PTL 2 discloses a system including at least two sets of superconducting logic devices, a cooling device adapted to cool the logic devices to a first operating temperature, and an interconnection coupling the superconducting logic devices.

[0006] PTL 3 discloses a quantum bit array including: a semiconductor layer; an insulating layer provided on the semiconductor layer; and a plurality of first gate electrodes provided on the insulating layer and configured to trap electrons in a predetermined spin state in the semiconductor layer by applying a voltage. The quantum bit array has a means to flow, in an extending direction of the first gate electrode, a current for forming a magnetic field acting on the electrons to at least one of the first gate electrodes when changing the spin state of the electrons.

CITATION LIST

Patent Literature

[0007] PTL 1: JP2013-090358A

[0008] PTL 2: JP2021-523572A

[0009] PTL 3: JP2022-130893A

SUMMARY OF THE INVENTION

[0010] For example, in order to arrange the quantum bit array disclosed in PTL 3 at the coolest portion (extremely low temperature environment) of a dilution refrigerator and control quantum computing at high speed, it is necessary to supply a synchronization signal with high accuracy. Since the space of a region maintained in the extremely low temperature environment is restricted, it is preferable to supply the synchronization signal from the outside of the extremely low temperature environment. However, it is difficult to reduce the wiring length in providing the synchronization signal from the outside, and it is difficult to improve the synchronization accuracy.

[0011] PTL 1 discloses a technique regarding system voltage detection and phase synchronization control in a series multiplex inverter device that operates in cooperation with a system. According to the technique, a phase synchronization signal is corrected (advanced) on the master control device side by a time required for generation thereof, thereby improving synchronization accuracy for a system of a plurality of slave control devices controlled by the signal. When a synchronization target serving as a reference of a device operation has a repetitive waveform of a system voltage or the like and a plurality of slave control devices perform the same operation, it can be expected that operations of all the slave control devices are efficiently synchronized with the system by using the technique.

[0012] A quantum computer having a quantum computing device as a core is an analog computer in which analog information is actually assigned to quantum dots. In order to obtain a desired fidelity of a quantum operation for determining final computing accuracy of a quantum computer, it is not sufficient to introduce an instrumentation device such as a bias voltage generation device or a radio-frequency generation device having necessary and sufficient accuracy and characteristics, or a minute voltage/current amplifying device used for reading a state of a quantum dot such as an electron spin, and to connect the instrumentation device to a quantum computing device. It is required to control an operation timing of each instrumentation device in accordance with a control sequence of a quantum operation desired to be executed, and to match (synchronize) relative operation timings between the devices with high time accuracy (for example, 10 nanoseconds).

[0013] Normally, the vendor of the introduced instrumentation device is not single, and in addition, a response delay time of the instrumentation device, that is, a time from when a device controlling the instrumentation device issues a predetermined operation request to when the requested operation is actually started on the instrumentation device side (including a propagation delay of the medium transmitting the operation request and a delay due to a protocol) is different for each instrumentation device. This characteristic is common even when a module measurement system of PXI (trademark) or the like is introduced, and it is needless to say that control synchronization between modules in consideration of a response delay time is necessary.

[0014] Therefore, an object of the invention is to provide an accurate synchronization signal for controlling quantum computing at high speed.

[0015] A preferred aspect of the invention is a quantum computing system including: a quantum device configured to execute a quantum operation; a plurality of instrumentation devices configured to transmit a control signal for executing the quantum operation to the quantum device; and a synchronization trigger distribution device configured to transmit a second trigger signal for instructing transmission of the control signal to at least one of the plurality of instrumentation devices based on a first trigger signal indicating a timing. The synchronization trigger distribution device includes a control sequence table for specifying a predetermined instrumentation device, to which the second trigger signal is to be output, among the plurality of instrumentation devices based on the first trigger signal, and a delay time table for specifying a delay time corresponding to each of the plurality of instrumentation devices. The synchronization trigger distribution device outputs the second trigger signal to the instrumentation device specified based on the control sequence table, at a timing based on the delay time specified based on the delay time table.

[0016] Another preferred aspect of the invention is an inter-device synchronization control method for quantum computing, the method using a quantum device configured to execute a quantum operation, a plurality of instrumentation devices configured to transmit a control signal for executing the quantum operation to the quantum device, and a synchronization trigger distribution device configured to transmit a second trigger signal for instructing transmission of the control signal to at least one of the plurality of instrumentation devices based on a first trigger signal indicating a timing. The method includes using the synchronization trigger distribution device to execute: a first step of specifying a predetermined instrumentation device, to which the second trigger signal is to be input, among the plurality of instrumentation devices based on the first trigger signal; a second step of specifying a delay time corresponding to each of the plurality of instrumentation devices; and a third step of outputting the second trigger signal to the instrumentation device specified based on the first step, at a timing based on the delay time specified based on the second step.

[0017] It is possible to supply an accurate synchronization signal for controlling the quantum computing at high speed.

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] FIG. **1** is a block diagram illustrating a configuration of a quantum computing system **1** according to a first embodiment;

[0019] FIG. **2** is a block diagram illustrating a configuration of a synchronization trigger distribution device **200**;

[0020] FIG. **3** is a table illustrating an example of settings in a control sequence table **230**;

[0021] FIG. **4** is a table illustrating an example of settings in a response delay time table **240**;

[0022] FIG. **5** is a table illustrating an example of definition of a command format of a master trigger signal **303**;

[0023] FIG. **6** is a timing chart of a control sequence of quantum operations in the quantum computing system **1**;

[0024] FIG. **7** is a detailed timing chart relating to sequence switching in the synchronization trigger distribution device **200**;

[0025] FIG. **8** is a block diagram illustrating a configuration of a quantum computing system **1**A according to a second embodiment;

[0026] FIG. **9** is a block diagram illustrating a configuration of a synchronization trigger distribution device **200**A;

[0027] FIG. **10** is a table illustrating an example of settings in a control sequence table **230**A;

[0028] FIG. **11** is a table illustrating an example of settings in a response delay time table **240**A;

[0029] FIG. **12** is a timing chart of a control sequence of quantum operations in the quantum computing system **1**A;

[0030] FIG. **13** is a block diagram illustrating a configuration of a quantum computing system **1**B according to a third embodiment; a block diagram illustrating a configuration of a quantum computing system **1**B according to a third embodiment (OMITTED)

[0031] FIG. **14** is a block diagram illustrating a configuration of a synchronization trigger distribution device **200**B;

[0032] FIG. **15** is a table illustrating an example of settings in a control sequence table **230**B;

[0033] FIG. **16** is a timing chart of a control sequence of quantum operations in the quantum computing system **1**B; and

[0034] FIG. **17** is a conceptual diagram in which a quantum computing system according to an embodiment is mounted in a dilution refrigerator.

DESCRIPTION OF EMBODIMENTS

[0035] Hereinafter, embodiments will be described with reference to the accompanying drawings. In the accompanying drawings, components functionally the same may be indicated by the same numbers or corresponding numbers. Although the accompanying drawings illustrate embodiments and examples in accordance with the principle of the present disclosure, the accompanying drawings are for understanding the present disclosure and are not intended to interpret the present disclosure in a limited manner. The description of the present specification is merely a typical example, and does not limit the scope of claims or application examples of the present disclosure in any sense.

[0036] In the embodiments, description is made in detail sufficiently in order for a person skilled in the art to implement the present disclosure, and other implementation and forms are also possible, and it is necessary to understand that a change in configuration and structure and replacement of various components are possible without departing from the scope and spirit of the technical idea of the present disclosure. Therefore, the following description should not be interpreted as being limited thereto.

[0037] In the configurations of the embodiments described below, the same portions or portions having the same functions are denoted by the same reference sign in different drawings, and redundant description thereof may be omitted.

[0038] When there are a plurality of components having the same or similar functions, the description may be made by assigning the same reference signs thereof with different subscripts. When it is not necessary to distinguish the plurality of components from each other, the description may be made by omitting the subscripts.

[0039] The notations "first", "second", "third", and the like in the present specification and the like are provided to identify the components, and do not necessarily limit the number, the order, or the content thereof. A number for identifying a component is used for each context, and the number used in one context does not necessarily indicate the same configuration in another context. In addition, this does not prevent a component identified by a certain number from also having a function of a component identified by another number.

[0040] In order to facilitate understanding of the invention, the position, size, shape, range, and the like of each configuration shown in the drawings or the like may not represent the actual position, size, shape, range, or the like. Therefore, the invention is not necessarily limited to the positions, sizes, shapes, ranges, or the like disclosed in the drawings.

[0041] Publications, patents, and patent applications cited in the present specification constitute a part of the description of the present specification.

[0042] In the present specification, a component represented in a single form includes a plurality of forms unless otherwise clearly described in the context.

[0043] In the embodiment, in relation to a quantum computer system implemented by combining a plurality of instrumentation devices having different response delay times, a configuration will be described which provides an inter-device synchronization control method necessary for implementing a control sequence of a desired quantum operation and achieving a fidelity requirement, a synchronization trigger distribution device which generates a trigger signal specific to the instrumentation device for which the response delay time is corrected according to a preset control sequence, and a quantum computer system including the synchronization trigger distribution device.

[0044] A quantum computing system according to an embodiment includes: a plurality of instrumentation devices configured to execute at least a part of quantum operations in a control sequence of the quantum operations; and a synchronization trigger distribution device configured to output a synchronization signal to the plurality of instrumentation devices. The synchronization trigger distribution device receives a trigger signal from a first instrumentation device. Based on a control sequence table for specifying whether it is a timing at which the synchronization signal is to be output to the plurality of instrumentation devices (including the first instrumentation device), and a delay time table for defining a processing delay time of the plurality of instrumentation devices starting from the synchronization signal, the synchronization trigger distribution device outputs a corrected synchronization signal whose timing is corrected so as to correct a specific processing delay time to the instrumentation device that is at a timing at which the synchronization signal is to be output.

[0045] Further, a modification of the synchronization trigger distribution device includes an execution time control unit that internally generates a signal in replace of the trigger signal received from the first instrumentation device and that can implement an autonomous control sequence independent of the trigger signal.

[0046] According to the inter-device synchronization control method, the instrumentation devices, and the synchronization trigger distribution device according to the embodiment, a highly practical quantum computing system is implemented in which operations of a plurality of instrumentation devices having different processing delay times are synchronized with high timing accuracy, and a control sequence of a desired quantum operation can be autonomously executed.

First Embodiment

[0047] FIG. **1** illustrates a configuration of a quantum computing system **1** according to a first embodiment of the invention. The embodiment is a system obtained by combining a host PC **100**, a plurality of instrumentation devices **300**, **400**, **410**, and **420**, and a synchronization trigger distribution device **200** to a quantum device **500** that executes a quantum operation. The host PC **100** has a system setting function **1001** for setting and managing the entire system. The plurality of instrumentation devices **300**, **400**, **410**, and **420** each execute at least a part of quantum operations in a control sequence of quantum operations. The synchronization trigger distribution device **200** outputs unique synchronization signals to the instrumentation devices **300**, **400**, **410**, and **420** respectively based on a control sequence table and a response delay time table set by the host PC **100**.

[0048] In the embodiment, the instrumentation device **300** and the synchronization trigger distribution device **200** have mutually consistent sequence control functions **3001** and **2001**, and the instrumentation device **300** serves as a sequence control master device and performs sequence control in cooperation with the synchronization trigger distribution device **200**.

[0049] The host PC **100** is usually installed in a room temperature environment (for example, 20° C.). On the other hand, a semiconductor chip constituting the quantum device **500** is disposed in an extremely low temperature environment (for example, −272° C.) realized by a mixing chamber whose temperature is the lowest in a cooling device such as a dilution refrigerator, and can operate a quantum dot. The quantum device **500** is similar to a configuration including the quantum bit array disclosed in PTL **3**, for example.

[0050] Other configurations are generally arranged in a room temperature environment, but among the instrumentation devices **300**, **400**, **410**, and **420**, the instrumentation device having the mode of a semiconductor chip may be set to an environment lower in temperature than the room temperature environment in the dilution refrigerator (however, not an extremely low temperature environment in which quantum dots operate). When the cooling capacity of the mixing chamber is sufficiently high, the instrumentation device having the mode of a semiconductor chip and the quantum device **500** may be implemented by the same semiconductor chip, and the entire device may be provided in an extremely low temperature environment.

[0051] The host PC **100** is a general information processing apparatus including a microprocessor, a memory, a storage device, an input device, an output device, and the like (not shown). The host PC **100** has the system setting function **1001** for performing device-specific settings for the synchronization trigger distribution device **200** and the instrumentation devices **300**, **400**, **410**, and **420**. In the system setting function **1001**, it is assumed that in at least some of the instrumentation devices **300**, **400**, **410**, and **420**, a system control application is running that performs, as a part of a control sequence, transfer of a quantum dot state read by predetermined measurement processing to the host PC **100** and state monitoring and operation management of the quantum computing system **1**.

[0052] The system control application interprets a control sequence of a quantum operation specified by a user as an object to be executed, supplies device-specific setting information necessary for the quantum operation to the instrumentation devices **400**, **410**, **420**, and **300** and the synchronization trigger distribution device **200** via communication means **101**, **102**, **103**, **104**, and **105**, respectively, and performs predetermined communication such as transferring a quantum dot state read by the instrumentation device. Here, the communication means **101**, **102**, **103**, **104**, and **105** may be interfaces based on standard communication specifications such as USB (trademark), SPI (trademark), I2C (trademark), RS-232C, GP-IB, PCI, Express (trademark), and Ethernet (trademark), or may have unique specifications based on general-purpose input and output ports.

[0053] The synchronization trigger distribution device **200** refers to setting information (a control sequence table and a response delay time table described later) transferred from the host PC **100** via the communication means **105** in advance, updates an internal state of the synchronization trigger distribution device **200** in accordance with contents of an operation request indicated by a master trigger signal **303** that is received from the instrumentation device **300** in accordance with the progress of the control sequence being executed, and independently outputs distribution trigger signals **201-1**, **201-2**, **201-3**, and **201-4** for instructing operation timings to the instrumentation devices **300**, **400**, **410**, and **420**, respectively. Details of the configuration and operation will be described later with reference to other drawings.

[0054] The instrumentation device **300** includes a bias voltage generation function **4001**, a radio-frequency signal generation function **4101**, and a computing result read function **4201** which are directly related to control of the quantum operation in the quantum device **500**. In addition to the above, the instrumentation device **300** includes an integrated sequence control function **3001** for unifying, with one command string, both of local sequence control that defines operation necessity, operation parameters, operation timings, order, and the like of the built-in functions (in the sense of being closed to the instrumentation device **300**), and global sequence control that realizes inter-device synchronization with the instrumentation devices **400**, **410**, and **420** by transmitting and receiving trigger signals via the synchronization trigger distribution device **200** (in the sense of controlling other devices).

[0055] Although not particularly limited, in order to implement the global sequence control, a command set supported by the sequence control function **3001** may include a master trigger output command for controlling output contents to the master trigger signal **303** and a distribution trigger standby command for standby until distribution trigger signal input to the own device via the distribution trigger signal **201-1**.

[0056] The master trigger output command includes a command for outputting each of four types of master trigger signals shown in FIG. **5** later, and when the instrumentation device **300** executes a command for outputting a master trigger signal, the master trigger signal corresponding to the command is transmitted to the synchronization trigger distribution device **200**.

[0057] Further, in order to recover from a stack state of control sequence processing, it is preferable to set, in the distribution trigger standby command, a timeout period for cancelling distribution trigger signal input standby based on the execution of the command. The distribution trigger standby command is a command for standby for a trigger signal to return from the synchronization trigger distribution device **200**, and can specify a timeout period as an execution parameter. The instrumentation device **300** has a timeout detecting function for detecting that a trigger signal input standby state continues for the specified timeout period or longer after the distribution trigger standby command is executed.

[0058] Here, the global sequence control focuses on an event-driven operation. That is, the sequence control master device (the instrumentation device **300** in the embodiment) that manages the overall sequence control of the quantum operation generates the master trigger signal **303** indicating timings of various events such as start and end of the control sequence, change in an operating state (during operation or in idle) of the instrumentation device, change in an output state of the instrumentation devices **400** and **410** that are in charge of control signal output, and measurement processing of the instrumentation device **420** that reads a quantum dot state.

[0059] The synchronization trigger distribution device **200** that receives the master trigger signal **303** outputs, only to an instrumentation device to be operated next in response to the event, a trigger signal that enables the operation of the instrumentation device among the distribution trigger signals **201-1**, **201-2**, **201-3**, and **201-4**.

[0060] Although not particularly limited, the instrumentation device **300** may not have a bias voltage generation function **4001-1**, a radio-frequency signal generation function **4101-1**, and a computing result read function **4201-1** which are directly related to the control of the quantum operation. In this case, the instrumentation device **300** may be handled as a sequence control dedicated device having only the sequence control function **3001** corresponding to the global sequence control.

[0061] Hereinafter, in the control sequence of the quantum operation, a section interposed between two temporally adjacent events is referred to as a subsequence, and it is assumed that the entire control sequence includes one or more subsequences. The sequence control function of the sequence control master device (in the embodiment, the instrumentation device **300**) is required to be controlled to explicitly secure a processing time for an instrumentation device other than the own device even in a subsequence in which the own device is not involved in input and output with the quantum device **500**. Therefore, it is preferable to define a command capable of describing a "processing standby" subsequence based on a "no-operation" operation capable of specifying an execution time.

[0062] The global and local sequence control operations of the instrumentation device **300** are based on the following flow. It is assumed that a command string necessary for control is transferred in advance from the host PC **100** via the communication means **104**, and the sequence control function **3001** of the instrumentation device **300** holds a command to be processed next in the command string or holds pointer information specifying a first subsequence, and stands by for execution until the input of the distribution trigger signal **201-1**.

[0063] Upon receiving the distribution trigger signal **201-1** from the synchronization trigger distribution device **200**, the sequence control function **3001** refers to the pointer information and sequentially processes a command string included in the subsequence. Specifically, if a control object of the command is the bias voltage generation function **4001-1**, a bias voltage control output signal **301** is generated according to a specified operation parameter, and if the control object is the radio-frequency signal generation function **4101-1**, a radio-frequency output signal **302** is generated. If the control object is the computing result read function **4201-1**, the local sequence control is implemented by measuring, by a predetermined means, at least one signal state included in a quantum computing unit output signal **521** output from the quantum device **500** and acquiring a quantum dot state.

[0064] At the end of the subsequence, the master trigger output command and the distribution trigger standby command are successively arranged, and by executing these commands, the global sequence control, that is, the master trigger signal **303** having a format corresponding to an execution request command of the next subsequence is output, and after the pointer information is updated, a series of processing up to transition to an input standby state of the distribution trigger signal **201-1** is performed.

[0065] As a part of the control sequence of the quantum operation, the instrumentation device **400** takes charge of, for example, only a bias voltage generation function **4001-2** for controlling a potential barrier, and generates a bias voltage control output signal **401** at a timing, at which the distribution trigger signal **201-2** is received, based on an operation parameter set in advance via the communication means **101** from the host PC **100**. After the operation is completed, standby is executed until the next distribution trigger signal is received.

[0066] As a part of the control sequence of the quantum operation, the instrumentation device **410** takes charge of, for example, only a radio-frequency signal generation function **4101-2** for state control of an electron spin, and generates a radio-frequency output signal **411** at a timing, at which the distribution trigger signal **201-3** is received, based on an operation parameter set in advance via the communication means **102** from the host PC **100**. After the operation is completed, standby is executed until the next distribution trigger signal is received.

[0067] As a part of the control sequence of the quantum operation, the instrumentation device **420** takes charge of, for example, only a computing result read function **4201-2**, and measures at least one signal state included in the quantum computing unit output signal **521** by a predetermined means at a timing, at which the distribution trigger signal **201-4** is received, based on an operation parameter set in advance via the communication means **103** from the host PC **100** and acquires the signal state as a quantum dot state. After the operation is completed, standby is executed until the next distribution trigger signal is received.

[0068] When the instrumentation device **300** has the bias voltage generation function **4001-1**, the radio-frequency signal generation function **4101-1**, and the computing result read function **4201-1**, and the instrumentation devices **400**, **410**, and **420** also have the bias voltage generation function **4001-2**, the radio-frequency signal generation function **4101-2**, and the computing result read function **4201-2**, respectively, the function of which instrumentation device is to be used may be selected based on a required specification.

[0069] The quantum device **500** includes a quantum computing control unit **510** and a quantum computing unit **520**.

[0070] The quantum computing control unit **510** is connected to at least one of the bias voltage control output signal **301** and the bias voltage control output signal **401** and temporarily stores at least a part of control information input from the bias voltage control output signal, switches a bias voltage application state of a quantum computing unit control signal **511** including a bias voltage control line, which is a direct control signal for each quantum dot of the quantum computing unit **520**, based on the stored control information and a strobe signal included in the bias voltage control output signal, and outputs the bias voltage application state to the quantum computing unit **520**.

[0071] The quantum computing unit **520** is a quantum dot array in which a plurality of quantum dots whose characteristics are variable by a bias voltage to be applied are connected in a one-dimensional or two-dimensional shape via a transfer gate for controlling the degree of electron transfer or interaction between the quantum dots by the bias voltage. In order to implement a desired quantum operation, the quantum computing unit **520** is connected to at least one of the radio-frequency output signal **302** and the radio-frequency output signal **411** in addition to the quantum computing unit control signal **511**. The quantum operation based on a bias voltage of the quantum computing unit control signal **511** and an application state of the radio-frequency output signal is executed, and the quantum computing unit output signal **521** including the quantum dot state is output.

[0072] FIG. **2** is a block diagram illustrating a detailed configuration of the synchronization trigger distribution device **200**. The synchronization trigger distribution device **200** includes a communication interface section **210**, a table read control unit **220**, a control sequence table **230**, a response delay time table **240**, and delay correction units **250-1**, **250-2**, **250-3**, and **250-4** equal in number to the distribution trigger signals **201-1**, **201-2**, **201-3**, and **201-4**. Details of the control sequence table **230** and the response delay time table **240** will be described later with reference to FIG. **3** and FIG. **4**, respectively.

[0073] The communication interface section **210** controls communication with the host PC **100** connected by the communication means **105**, and mutually converts various control signals among an internal communication means **211** for controlling access to the table read control unit **220**, an internal communication means **212** for controlling access to the control sequence table **230**, and an internal communication means **213** for controlling access to the response delay time table **240**.

[0074] The table read control unit **220** counts the number of times of reception of the master trigger signal **303** (MREQ) by an internal trigger number counter **2201**, and specifies a subsequence being executed in the control sequence of the quantum operation based on the count value. In addition, in synchronization with the reception timing of the master trigger signal **303**, the table read control unit **220** outputs a read control signal **221** including read entry position information in the control sequence table **230** and a read request to the control sequence table **230**, and a timer start request signal **222** for a timer function for time measurement provided in each of the delay correction units **250-1**, **250-2**, **250-3**, and **250-4**. The trigger number counter **2201** is designed such that a setting thereof can be updated from the host PC **100** via the communication means **105** and the internal communication means **211**. Although not particularly limited, the timer start request signal **222** may include an initial value commonly set for the timer functions of all the delay correction units at the time of requiring timer start.

[0075] The table read control unit **220** receives an auxiliary operation control signal **232** from the control sequence table **230** as a response to the read control signal **221**, and executes processing based on request contents.

[0076] In the control sequence table **230**, subsequence setting information from an entry position, which is specified by the read control signal **221**, in the table based on the read request from the read control signal **221**. In the read subsequence setting information, a timer enable signal **231-1** indicating "enabled or disabled" of the timer function of the delay correction unit **250-1** that corrects an output timing of the distribution trigger signal **201-1** (DREQ0) is output to the delay correction unit **250-1**, a timer enable signal **231-2** indicating "enabled or disabled" of the timer function of the delay correction unit **250-2** that corrects an output timing of the distribution trigger signal **201-2** (DREQ1) is output to the delay correction unit **250-2**, a timer enable signal **231-3** indicating "enabled or disabled" of the timer function of the delay correction unit **250-3** that corrects an output timing of the distribution trigger signal **201-3** (DREQ2) is output to the delay correction unit **250-3**, a timer enable signal **231-4** indicating "enabled or disabled" of the timer function of the delay correction unit **250-4** that corrects an output timing of the distribution trigger signal **201-4** (DREQ3) is output to the delay correction unit **250-4**, and the auxiliary operation control signal **232** indicating processing to be executed by the table read control unit **220** is output to the table read control unit **220** in synchronization with the reception timing of the master trigger signal **303**.

[0077] In the control sequence table **230**, it is assumed that setting information necessary for a control sequence of a quantum operation desired to be implemented is set before the start of execution of the control sequence from the host PC **100** via the communication means **105** and the internal communication means **212**. Further, it is assumed that the system control application operating on the host PC **100** is responsible for guaranteeing that there is no inconsistency with the sequence control setting information on the instrumentation device **300** side.

[0078] The response delay time table **240** holds information on a response delay time of the instrumentation devices **300**, **400**, **410**, and **420** measured in advance, that is, information on a time elapsed from when a device controlling the instrumentation device issues a predetermined operation request to when the requested operation is actually started on the instrumentation device side, and outputs the information as delay correction values **241-1**, **241-2**, **241-3**, and **241-4** to the delay correction units **250-1**, **250-2**, **250-3**, and **250-4**, respectively.

[0079] In the response delay time table **240**, it is assumed that the response delay time of each instrumentation device is set before the start of the execution of the control sequence from the host PC **100** via the communication means **105** and the internal communication means **213**.

[0080] The response delay times of the instrumentation devices **300**, **400**, **410**, and **420** are preferably measured in a state where the entire system including connections between the devices is mounted. As an example of the measurement method, each instrumentation device may be set to operate so that the control output signal of the instrumentation device changes stepwise, such as on or off, asserted or negated, high level or low level, in response to the input of the distribution trigger signal, and the subsequence may be switched in a state in which the response delay time of all instrumentation devices is designated as zero in the response delay time table **240**. By measuring the elapse of time until a change point in the output state of each instrumentation device based on a timing of a subsequence switching request by the master trigger signal **303**, the response delay time of each instrumentation device can be collectively acquired. The response delay time in this case corresponds to a delay time starting from the distribution trigger signal. The above procedure may be manually executed, or may be automatically executed as a part of the functions of the system control application operating on the host PC **100**.

[0081] The delay correction unit **250-1** for correcting the output timing of the distribution trigger signal **201-1** (DREQ0) has a timer function including a time measurement counter. When the timer enable signal **231-1** is enabled and the timer start request signal **222** indicates that the start request is valid, the time measurement counter continuously performs a decrement operation based on a clock signal input (not shown) with a predetermined value such as an initial value included in the timer start request signal **222** as a count start value. When it is detected that a time indicated by the count value of the time measurement counter and a time indicated by the delay correction value **241-1** coincide with each other, that is, when it is detected that the correction of the response delay time is completed, a pulse signal having a fixed assertion time width is output as the distribution trigger signal **201-1** (DREQ0) to the instrumentation device **300**, and the decrement operation is stopped.

[0082] The delay correction unit **250-2** for correcting the output timing of the distribution trigger signal **201-2** (DREQ1) has a timer function including a time measurement counter. When the timer enable signal **231-2** is enabled and the timer start request signal **222** indicates that the start request is valid, the time measurement counter continuously performs a decrement operation based on a clock signal input (not shown) with a predetermined value such as an initial value included in the timer start request signal **222** as a count start value. When it is detected that a time indicated by the count value of the time measurement counter and a time indicated by the delay correction value **241-2** coincide with each other, that is, when it is detected that the correction of the response delay time is completed, a pulse signal having a fixed assertion time width is output as the distribution trigger signal **201-2** (DREQ1) to the instrumentation device **400**, and the decrement operation is stopped.

[0083] The delay correction unit **250-3** for correcting the output timing of the distribution trigger signal **201-3** (DREQ2) has a timer function including a time measurement counter. When the timer enable signal **231-3** is enabled and the timer start request signal **222** indicates that the start request is valid, the time measurement counter continuously performs a decrement operation based on a clock signal input (not shown) with a predetermined value such as an initial value included in the timer start request signal **222** as a count start value. When it is detected that a time indicated by the count value of the time measurement counter and a time indicated by the delay correction value **241-3** coincide with each other, that is, when it is detected that the correction of the response delay time is completed, a pulse signal having a fixed assertion time width is output as the distribution trigger signal **201-3** (DREQ2) to the instrumentation device **410**, and the decrement operation is stopped.

[0084] The delay correction unit **250-4** for correcting the output timing of the distribution trigger signal **201-4** (DREQ3) has a timer function including a time measurement counter. When the timer enable signal **231-4** is enabled and the timer start request signal **222** indicates that the start request is valid, the time measurement counter continuously performs a decrement operation based on a clock signal input (not shown) with a predetermined value such as an initial value included in the timer start request signal **222** as a count start value. When it is detected that a time indicated by the count value of the time measurement counter and a time indicated by the delay correction value **241-4** coincide with each other, that is, when it is detected that the correction of the response delay time is completed, a pulse signal having a fixed assertion time width is output as the distribution trigger the signal **201-4** (DREQ3) to instrumentation device **420**, and the decrement operation is stopped.

[0085] Here, for example, the time measurement counters in the delay correction units **250-1**, **250-2**, **250-3**, and **250-4** start the decrement operation from a common count start value, and stop the decrement operation independently at a time point when the correction of the response delay time is completed. By changing a stop condition of the decrement operation to a count value of zero, the operation of all the time measurement counters becomes common, so that the plurality of time measurement counters can be integrated into one and a hardware scale can be saved.

[0086] The count start values set in the time measurement counters in the delay correction units **250-1**, **250-2**, **250-3**, and **250-4** are main factors for determining time overheads associated with the subsequence switching, and the values should be chose with care. Although not particularly limited, any one of the following (1) to (3) may be used.

[0087] (1) Fixation to a maximum value allowed as the specification of the quantum computing system: the hardware scale related to the sequence control can be saved, but the computing performance of the system is extremely degraded when a switching time overhead cannot be ignored as compared with a time (coherence time) in which the quantum dot can maintain quantum properties.

[0088] (2) According to the response delay time of the instrumentation device having the maximum response delay time among instrumentation devices involved in one or more times of the control operation in the control sequence of the quantum operation desired to be executed: since one count start value can be statically determined in the entire control sequence, it is easy to balance an increase in hardware scale and the switching time overhead.

[0089] (3) According to the response delay time of the instrumentation device having the maximum response delay time among one or more instrumentation devices involved in the control operation in a subsequence being executed: the overhead associated with the switching is minimized by changing the count start value for each subsequence, but it requires a means for designating the count start value for each subsequence, for example, extension of the response delay time table **240**.

[0090] In addition, when determining the count start value in (2) or (3), it is necessary to consider a case where the response delay time of a specific instrumentation device is extremely long compared to those of other instrumentation devices. In this case, an upper limit is set to a value that can be set as the count start value, and with respect to an instrumentation device having a response delay time exceeding the upper limit, a difference response delay time obtained by subtracting the upper limit of the count start value from the response delay time of the instrumentation device may be separately handled. Specifically, the output timing of the distribution trigger signal is advanced by the difference response delay time by inserting an empty subsequence, which has an execution time corresponding to the difference response delay time and is configured to output the distribution trigger signal to the instrumentation device, immediately before the subsequence in which the instrumentation device originally operates.

[0091] The empty subsequence may be further divided into two or more subsequences in which the sum of the execution time and the overhead associated with the switching coincides with the difference response delay time. In the empty subsequence, or in a subsequence obtained by dividing the empty subsequence, an operation other than advancing the distribution trigger signal output may be requested to an instrumentation device other than the instrumentation device.

[0092] FIG. **3** illustrates an example of settings in the control sequence table **230**. The control sequence table includes a plurality of entries. Each entry identified by an entry number **2301** (corresponding to the count value of the trigger number counter **2201**) is associated with a specific subsequence.

[0093] One entry has distribution trigger signal output request fields **2302-1**, **2302-2**, **2302-3**, and **2302-4** for specifying whether the instrumentation devices **300**, **400**, **410**, and **420** should operate in the subsequence, that is, whether distribution trigger signal output is necessary for a predetermined instrumentation device **300**. Further, each entry has an auxiliary operation request field **2303** for specifying auxiliary operation contents to be processed by the table read control unit **220** in parallel with the execution of the subsequence.

[0094] When a read request is received from the table read control unit **220** via the read control signal **221**, setting information related to the subsequence is read from a specified entry position in the control sequence table **230**, contents of the distribution trigger signal output request fields **2302-1**, **2302-2**, **2302-3**, and **2302-4** are output as the timer enable signals **231-1**, **231-2**, **231-3**, and **231-4**, respectively, and contents of the auxiliary operation request field **2303** are output as the auxiliary operation control signal **232**.

[0095] In the control sequence table **230**, the distribution trigger signal to be output is specified, and since the distribution trigger signal corresponds to the instrumentation device, it is synonymous with specifying the instrumentation device corresponding to the count value of the trigger number counter **2201** by the control sequence table **230**.

[0096] FIG. **4** illustrates an example of settings in the response delay time table **240**. The response delay time table includes entries the number of which is equal to the number of trigger signal types **2401**, that is, the number of the instrumentation devices **300**, **400**, **410**, and **420**, and each entry has a response delay time field **2402** for specifying a response delay time measured in advance in relation to the instrumentation devices **300**, **400**, **410**, and **420** to which the distribution trigger signals **201-1**, **201-2**, **201-3**, and **201-4** are connected.

[0097] The delay response time is associated with the distribution trigger signal in the response delay time table **240**, and since the distribution trigger signal corresponds to the instrumentation device, it is synonymous with specifying the response delay time corresponding to each of the plurality of instrumentation devices by the response delay time table **240**.

[0098] The response delay time is, for example, an actual measurement value obtained in advance, and reflects at least one of a propagation delay of a transmission medium connecting the plurality of instrumentation devices and a delay based on a communication protocol. As described above, when the wiring between the blocks reaches, for example, several meters by arranging the blocks (for example, the configurations in FIG. **1**) of the device in the cooling device, it is necessary to sufficiently consider the influence of the response delay time.

[0099] FIG. **5** illustrates an example of definition of a command format of the master trigger signal **303**. In order to execute a consistent control sequence of quantum operations in the quantum computing system **1**, it is assumed that, in addition to a fact that control information set in advance does not have a defect, the operation of the sequence control function **3001** of the instrumentation device **300** and the operation of the table read control unit **220** of the synchronization trigger distribution device **200** have coincident subsequences recognized as being executed by a request/response protocol via the master trigger signal **303** and the distribution trigger signal **201-1**, that is, the sequence control synchronization is established.

[0100] For example, even under conditions that the synchronization trigger distribution device **200** is provided in a room temperature environment outside the dilution refrigerator and the instrumentation device **300** is provided in a low temperature environment inside the dilution refrigerator, leading to an increase in the wiring length of the medium responsible for transmission of the trigger signals and a deterioration in an electromagnetic failure level received by the medium, it is desirable to design a protocol capable of securing certain reliability, such as defining a command format which is not a simple single pulse particularly for the request side (master trigger signal) and providing a unit detecting that sequence control synchronization is not established at least.

[0101] Four types of master trigger output commands are defined, and an operation outline of each command is as follows. In the sequence control function **3001** of the instrumentation device **300**, the master trigger signal **303**, which is seamlessly interpreted and processed together with a command for controlling the built-in functions and has a format defined for each command, is output from the instrumentation device **300**.

SETCNTR:

[0102] This command requests the table read control unit **220** to update the count value of the trigger number counter **2201**, which is associated with the subsequence to be executed next and is corresponding to the entry number **2301** in the control sequence table **230**, to a specified value. Normally, before the start of execution of the control sequence, that is, when an operating state of the table read control unit **220** is a sequence non-execution state, this command is executed as a part of preprocessing of the quantum operation in order to specify a first subsequence of the control sequence to be executed. When a response from the synchronization trigger distribution device **200** is necessary or unnecessary after the execution of this command, the setting of the distribution trigger signal output request field **2302-1** of the entry in the control sequence table **230** specified by this command is set to "valid" or "invalid" correspondingly.

[0103] The table read control unit **220**, which receives the master trigger signal **303** corresponding to this command, updates the count value of the built-in trigger number counter **2201** to the count value notified by this command.

CHKCNTR:

[0104] This command is a command for the instrumentation device **300** to notify the synchronization trigger distribution device **200** of information indicating its own processing that is, information specifying which subsequence is currently being executed, for the purpose of checking whether a synchronization deviation of the processing occurs. The sequence control function of the instrumentation device **300** notifies the table read control unit **220** of the count value of the trigger number counter **2201**, which is associated with the subsequence recognized as being executed at the time of execution of the command and is corresponding to the entry number **2301** in the control sequence table **230**.

[0105] Normally, during execution of the control sequence, that is, when the operating state of the table read control unit **220** is a sequence execution state, this command is inserted into the control sequence non-periodically for the purpose of confirming that there is no synchronization deviation in the sequence control between the instrumentation device **300** and the synchronization trigger distribution device **200**. A setting related to response necessity after the execution of this command is common to that of the SETCNTR command.

[0106] The table read control unit **220**, which receives the master trigger signal **303** corresponding to this command, confirms whether the count value of the built-in trigger number counter **2201** coincides with the count value notified by this command. If the result is that the count values coincide with each other, the host PC **100** is notified via the internal communication means **211** and the communication means **105** that sequence control synchronization is not established.

NEXT:

[0107] This command requests the table read control unit **220** to increment the setting of the count value of the trigger number counter **2201**, which is associated with the subsequence to be executed next and is corresponding to the entry number **2301** in the control sequence table **230**, that is, to execute the subsequent subsequence. Although a setting related to response necessity after the execution of this command is common to that of the SETCNTR command, it should be set as "response necessary" in order to maintain a sequence control synchronous state.

[0108] The table read control unit **220**, which receives the master trigger signal **303** corresponding to this command, increments the count value of the built-in trigger number counter **2201** and sets the operating state to the sequence execution state if the operating state is the sequence non-execution state.

END:

[0109] Similarly to the NEXT command, this command requests the table read control unit **220** to increment the setting of the count value of the trigger number counter **2201**. Normally, this command is used for the purpose of requesting execution of postprocessing following the quantum operation. Although a setting related to response necessity after the execution of this command is common to that of the SETCNTR command, it should be set as "response necessary" in order to maintain a sequence control synchronous state.

[0110] The table read control unit **220**, which receives the master trigger signal **303** corresponding to this command, increments the count value of the built-in trigger number counter **2201** and sets the operating state to the sequence non-execution state if the operating state is the sequence execution state.

[0111] The command configuration described above is an example. For example, the NEXT command may be omitted, and a simple timing pulse may be counted by a counter on the receiving side. At this time, commands for instructing count start and count end of the counter are required.

[0112] The CHKCNTR command may request not only the instrumentation device **300** but also other instrumentation devices **400**, **410**, and **420** to notify the synchronization trigger distribution device **200** of information indicating their processing states, that is, information specifying which subsequence is currently being executed.

[0113] FIG. **6** is a time chart schematically illustrating a flow of processing when a control sequence of the quantum operation based on the settings in the control sequence table **230** illustrated in FIG. **3** is executed in the quantum computing system **1** according to the first embodiment.

[0114] Symbols "S", "N", and "E" written together with the master trigger signal **303** in the drawing indicate outputs according to the SETCNTR command, the NEXT command, and the END command, respectively. A subsequence denoted by (Wait) in the operation of the instrumentation device **300** indicates a "standby state" in which no direct control output to the quantum device **500** is performed and only the execution time of the subsequence is managed.

[0115] The entire control sequence includes seven subsequences indicated by "preprocess" (trigger number counter=0), SEQ1 (trigger number counter=1), SEQ2 (trigger number counter=2), SEQ3 (trigger number counter=3), SEQ4 (trigger number counter=4), SEQ5 (trigger number counter=5), and "post-process" (trigger number counter=6). It is assumed that, prior to the start of execution of the control sequence, the contents of the control sequence table **230** and the response delay time table **240** are appropriately initialized by the system control application running on the host PC **100** via the internal communication means **212** and **213**, respectively.

[0116] When the initial setting is completed, the host PC **100** sets the trigger number counter **2201** in the table read control unit **220** to zero via the communication means **105** and the internal communication means **211** (in this example, the processing is not essential).

[0117] Subsequently, the host PC **100** instructs, via the communication means **104**, the sequence control function **3001** in the instrumentation device **300** to start execution from a first subsequence of the control sequence, that is, a preprocessing subsequence. The sequence control function **3001** in the instrumentation device **300**, which receives the instruction, executes the SETCNTR command as first processing in the preprocessing subsequence (sets so as not to wait for a response by the distribution trigger signal **201-1**), and requests to update the trigger number counter **2201** in the table read control unit **220** to zero via the master trigger signal **303**. Further, at the end of the preprocessing subsequence, the NEXT command is executed as second processing (setting is made so as to wait for a response by the distribution trigger signal **201-1**), execution of a subsequent SEQ1 subsequence is requested via the master trigger signal **303**, and a response from the synchronization trigger distribution device **200** is waited for according to the distribution trigger standby command (not shown).

[0118] The table read control unit **220**, which receives the request to execute the next subsequence from the master trigger signal **303**, shifts the operating state to the sequence execution state and performs increment in the trigger number counter **2201**. Further, the control sequence table **230** is requested to read setting information for the SEQ1 subsequence. When the timer enable signal **231-1** indicating "enabled or disabled" of the timer function in the SEQ1 subsequence read from the control sequence table **230** indicates "enabled", the delay correction unit **250-1** is requested, through the timer start request signal **222**, to perform delay correction processing necessary for the distribution trigger signal output by referring to the delay correction value **241-1** related to the instrumentation device **300** output from the response delay time table **240**.

[0119] The delay correction unit **250-1** which receives the request outputs the distribution trigger signal **201-1**, whose output timing is corrected based on the response delay time of the instrumentation device **300**, to the instrumentation device **300**, whereby the instrumentation device **300** starts execution of the SEQ1 subsequence. At the end of the SEQ1 subsequence, the NEXT command is executed again (setting is made so as to wait for a response by the distribution trigger signal **201-1**), execution of a subsequent SEQ2 subsequence is requested, and a response from the synchronization trigger distribution device **200** is waited for according to the trigger standby command (not shown).

[0120] By the above procedure, switching between execution target subsequences in cooperation between the synchronization trigger distribution device **200** and the instrumentation device **300** is completed. This switching is repeated up to an SEQ5 subsequence. Although operation control of the instrumentation devices **400**, **410**, and **420** is basically common to that of the instrumentation device **300**, the instrumentation devices **400**, **410**, and **420** are different from the instrumentation device **300** in that the instrumentation devices **400**, **410**, and **420** execute only preset operations in synchronization with the distribution trigger signals **201-2**, **201-3**, and **201-4** and do not contribute to the sequence switching control.

[0121] The END command is executed at the end of the SEQ5 subsequence (setting is made so as to wait for a response by the distribution trigger signal **201-1**), execution of the subsequent post-processing subsequence is requested via the master trigger signal **303**, and a response from the synchronization trigger distribution device **200** is waited for according to the distribution trigger standby command (not shown).

[0122] The table read control unit **220**, which receives the request to execute the next subsequence from the master trigger signal **303**, shifts the operating state to the sequence non-execution state and performs increment in the trigger number counter **2201**. Further, the control sequence table **230** is requested to read setting information for the post-processing subsequence. When the timer enable signal **231-1** indicating "enabled or disabled" of the timer function in the post-processing subsequence read from the control sequence table **230** indicates "enabled", the delay correction unit **250-1** is requested, through the timer start request signal **222**, to perform delay correction processing necessary for the distribution trigger signal output by referring to the delay correction value **241-1** related to the instrumentation device **300** output from the response delay time table **240**.

[0123] Based on the request of the auxiliary operation control signal **232**, the table read control unit **220** notifies the host PC **100** of the completion of the execution of the control sequence via the communication means **105** in parallel with the switching to the post-processing subsequence.

[0124] The delay correction unit **250-1** which receives the request outputs the distribution trigger signal **201-1**, whose output timing is corrected based on the response delay time of the instrumentation device **300**, to the instrumentation device **300**, whereby the instrumentation device **300** starts execution of the post-processing subsequence. After the execution of the post-processing subsequence is completed, an idle state is established until a new operation request is received from the host PC **100**.

[0125] FIG. **7** is a timing chart showing detailed processing in the synchronization trigger distribution device **200** in relation to execution switching from the SEQ2 subsequence to the SEQ3 subsequence, as an example, in the schematic timing chart of the quantum computing sequence described with reference to FIG. **6**.

[0126] The table read control unit **220**, which receives the request to execute the SEQ3 subsequence from the master trigger signal **303**, performs increment in the trigger number counter **2201**. The control sequence table **230** is requested to read setting information for the SEQ3 subsequence of the entry number **2301** (=3) via the read control signal **221** designating the incremented count value (=3) as the entry position information.

[0127] At a timing the setting information reading from the control sequence table **230** is completed, if the timer enable signal **231-1** indicating "enabled or disabled" of the timer function in the SEQ3 subsequence read from the control sequence table **230** indicates "enabled", the table read control unit **220** requests, through the timer start request signal **222**, the delay correction unit **250-1** to perform delay correction processing necessary for the distribution trigger signal output by referring to the delay correction value **241-1** related to the instrumentation device **300** output from the response delay time table **240**.

[0128] The delay correction unit **250-1**, which receives the request, enables the built-in timer function for time measurement, sets a timer initial value included in the timer start request signal **222** as an initial count value in the time measurement counter which is a component implementing the timer function, and then continuously performs a decrement operation of the timer in synchronization with a predetermined clock input signal. When it is detected that a time indicated by the count value of the time measurement counter and a time indicated by the delay correction value **241-1** coincide with each other, that is, when it is detected that the correction of the response delay time is completed, as a trigger signal of the response delay time correction completion, a pulse signal having a fixed assertion time width is output to the instrumentation device **300** as the distribution trigger signal **201-1**, and the decrement operation is stopped, and the timer function is disabled. The series of procedures are simultaneously performed for the delay correction units **250-2**, **250-3** (not shown), and **250-4** (not shown).

[0129] According to the first embodiment described in detail above with reference to the drawings, when the quantum computing system **1** including the plurality of instrumentation devices **300**, **400**, **410**, and **420** executes a control sequence of a quantum operation, the synchronization trigger distribution device **200** having a function of individually correcting a response delay time of each instrumentation device with a subsequence as the control unit is introduced, whereby operations of the instrumentation devices can be synchronized with high time accuracy with a relatively simple configuration, and the requirement related to fidelity of the quantum operation can be satisfied.

[0130] Further, regarding the instrumentation device **300** which is a master device of the sequence control in the quantum computing system **1**, in addition to a local sequence control command for controlling built-in functions, a command for supporting synchronization between the instrumentation devices and implementing global sequence control is defined, and these commands can be seamlessly handled in one command string, whereby it is possible to implement a sequence control function having high controllability and improve the practicability of the quantum computing system **1**.

Second Embodiment

[0131] A quantum computing system **1A** according to a second embodiment of the invention will be described with reference to FIG. **8**. Since the system configuration of the second embodiment is substantially the same as that of the first embodiment, redundant description will be omitted.

[0132] The configuration of the quantum computing system **1A** is different from the configuration of the quantum computing system **1** (FIG. **1**) in the following points. That is, a synchronization trigger distribution device **200**A replaces the synchronization trigger distribution device **200**, the master trigger signal **303** output by the sequence control function **3001** of the instrumentation device **300** is deleted, and a master trigger signal **110** and a distribution trigger signal **202** are newly added between the synchronization trigger distribution device **200** and the host PC **100**. According to the configuration change, the role of the master device in the control sequence of the quantum operation is transferred from the instrumentation device **300** to the host PC **100**, and the host PC **100** has a sequence control function **3001**A. On the other hand, the instrumentation device **300** executes only the bias voltage generation function **4001-1**, the radio-frequency signal generation function **4101-1**, and the computing result read function **4201-1**, which are preset local sequence control based on the distribution trigger signal **201-1**, similarly to the instrumentation devices **400**, **410**, and **420**.

[0133] FIG. **9** is a block diagram illustrating a detailed configuration of the synchronization trigger distribution device **200**A. Since the configuration of the synchronization trigger distribution device **200**A is substantially the same as the configuration of the synchronization trigger distribution device **200**, redundant description will be omitted.

[0134] The configuration of the synchronization trigger distribution device **200**A is different from the configuration of the synchronization trigger distribution device **200** in the following points. That is, the table read control unit **220** receives the master trigger signal **110** instead of the master trigger signal **303**, and outputs a trigger signal, in which a response delay time is corrected by a newly added delay correction unit **250-5**, as the distribution trigger signal **202** to the host PC **100**. The delay correction unit **250-5** has the same function as the delay correction units **250-1**, **250-2**, **250-3**, and **250-4**.

[0135] Here, the host PC **100** may manage at least an execution time (elapsed time from the start of execution) of a subsequence, and in addition, may perform processing (pre-processing and post-processing) during a period in which each instrumentation device is in an idle state, that is, an operating state of the table read control unit **220** indicates a sequence non-execution state. In any case, the instrumentation devices **300**, **400**, **410**, and **420** perform the control operation for operating the quantum device **500**.

[0136] Since the timer enable signal **233** indicating whether the timer function of the delay correction unit **250-5** is enabled or disabled can be output to the delay correction unit **250-5**, the control sequence table **230** is replaced with a control sequence table **230**A to be described later with reference to FIG. **10**, and since a delay correction value **241-5** indicating the response delay time of the host PC **100** can be output to the delay correction unit **250-5**, the response delay time table **240** is replaced with a response delay time table **240**A to be described later with reference to FIG. **11**.

[0137] FIG. **10** illustrates an example of settings in the control sequence table **230**A. The control sequence table includes a plurality of entries, and each entry identified by the entry number **2301** (corresponding to the count value of the trigger number counter **2201**) is associated with a specific subsequence. One entry includes distribution trigger signal output request fields **2302-1**, **2302-2**, **2302-3**, **2302-4**, and **2302-5** for specifying whether the instrumentation devices **300**, **400**, **410**, and **420** and the host PC **100** should operate in the subsequence, that is, whether distribution trigger signal output is necessary, and the auxiliary operation request field **2303** for specifying auxiliary operation contents to be processed by the table read control unit **220** in parallel with execution of the subsequence.

[0138] When a read request is received from the table read control unit **220** via the read control signal **221**, it is requested to read setting information related to the subsequence from a specified entry position in the control sequence table **230**A, contents of the distribution trigger signal output request fields **2302-1**, **2302-2**, **2302-3**, **2302-4**, and **2302-5** are output as the timer enable signals **231-1**, **231-2**, **231-3**, **231-4**, and **233** respectively, and contents of the auxiliary operation request field **2303** are output as the auxiliary operation control signal **232**.

[0139] FIG. **11** illustrates an example of settings in the response delay time table **240**A. The response delay time table includes entries the number of which is equal to the number of trigger signal types **2401**, that is, the number of the instrumentation devices **300**, **400**, **410**, **420** and the host PC **100**, and each entry has the response delay time field **2402** for specifying a response delay time measured in advance in relation to the instrumentation devices **300**, **400**, **410**, **420** and the host PC **100** to which the distribution trigger signals **201-1**, **201-2**, **201-3**, **201-4**, and **202** are connected.

[0140] FIG. **12** is a time chart schematically illustrating a flow of processing when a control sequence of the quantum operation based on the settings in the control sequence table **230**A illustrated in FIG. **10** is executed in the quantum computing system **1**A according to the second embodiment. A main difference from the time chart in the first embodiment is that the host PC **100** performs preprocessing, postprocessing, and execution time management of each subsequence via the master trigger signal **110** and the distribution trigger signal **202**.

[0141] When the initial setting (not shown) of the control sequence table **230**A and the response delay time table **240**A is completed, the host PC **100** sets the trigger number counter **2201** in the table read control unit **220** to zero via the communication means **105** and the internal communication means **211**.

[0142] Subsequently, the host PC **100** executes a predetermined preprocessing subsequence, further executes the NEXT command at the end of the preprocessing subsequence (sets to wait for a response by the distribution trigger signal **202**), requests execution of the subsequent SEQ1 subsequence via the master trigger signal **110**, and waits for a response from the synchronization trigger distribution device **200**.

[0143] The table read control unit **220**, which receives the request to execute the next subsequence from the master trigger signal **110**, shifts the operating state to the sequence execution state and performs increment in the trigger number counter **2201**. Further, the control sequence table **230**A is requested to read setting information for the SEQ1 subsequence. When the timer enable signal **233** indicating "enabled or disabled" of the timer function in the SEQ1 subsequence read from the control sequence table **230**A indicates "enabled", the delay correction unit **250-5** is requested, via the timer start request signal **222**, to perform delay correction processing necessary for the distribution trigger signal output by referring to the delay correction value **241-5** related to the host PC **100** output from the response delay time table **240**A.

[0144] The delay correction unit **250-5** which receives the request outputs the distribution trigger signal **202**, whose output timing is corrected based on the response delay time of the host PC **100**, to the host PC **100**, whereby the host PC **100** starts execution of the processing included in the SEQ1 subsequence and execution time management of the subsequence. At the end of the SEQ1 subsequence processing, the NEXT command is executed again (setting is made so as to wait for a response by the distribution trigger signal **202**), execution of the subsequent SEQ2 subsequence is requested, and a response from the synchronization trigger distribution device **200** is waited for again.

[0145] By the above procedure, switching between execution target subsequences in cooperation between the synchronization trigger distribution device **200** and the host PC **100** is completed. This switching is repeated up to the SEQ5 subsequence.

[0146] The host PC **100** executes the END command at the end of the SEQ5 subsequence processing (sets so as to wait for a response by the distribution trigger signal **202**), requests execution of the subsequent post-processing subsequence via the master trigger signal **110**, and waits for a response from the synchronization trigger distribution device **200**.

[0147] The table read control unit **220**, which receives the request to execute the next subsequence from the master trigger signal **110**, shifts the operating state to the sequence non-execution state and performs increment in the trigger number counter **2201**. Further, the control sequence table **230**A is requested to read setting information for the post-processing subsequence. When the timer enable signal **233** indicating "enabled or disabled" of the timer function in the post-processing subsequence read from the control sequence table **230**A indicates "enabled", the delay correction unit **250-5** is requested, via the timer start request signal **222**, to perform delay correction processing necessary for the distribution trigger signal output by referring to the delay correction value **241-5** related to the host PC **100** output from the response delay time table **240**A.

[0148] The delay correction unit **250-5** which receives the request outputs the distribution trigger signal **202**, whose output timing is corrected based on the response delay time of the host PC **100**, to the host PC **100**, whereby the host PC **100** starts execution of the post-processing subsequence. The host PC **100** makes a notification (not shown) to the system control application that the control sequence of the quantum operation is completed, for example, as partial processing of the post-processing subsequence, and stands by in an idle state until the next processing request.

[0149] In the second embodiment, if time resolution required for the execution time measurement of the subsequence is at a level that can be realized by a timer function of an operating system running on the host PC **100**, the execution time management can be transferred to the host PC **100**. As a result, a master trigger signal output function is not required on the instrumentation device side, and the instrumentation device can be replaced with a more inexpensive instrumentation device.

Third Embodiment

[0150] A quantum computing system **1**B according to a third embodiment of the invention will be described with reference to FIG. **13**. Since the system configuration of the third embodiment is substantially the same as those of the first and second embodiments, redundant description will be omitted.

[0151] The configuration of the quantum computing system **1**B is different from the configuration of the quantum computing system **1**A (FIG. **8**) in the following points. That is, in the synchronization trigger distribution device **200**, the sequence control function **3001**A of the host PC **100** in the second embodiment is replaced by a synchronization trigger distribution device **200**B to which an execution time management function is added, and the master trigger signal **110** output from the host PC **100** is deleted. According to the configuration change, the role of the master device in the control sequence of the quantum operation is transferred from the instrumentation device **300** to the synchronization trigger distribution device **200**B, and the instrumentation device **300** executes only preset local sequence control based on the distribution trigger signal **201-1** similarly to the instrumentation devices **400**, **410**, and **420**. It is assumed that only the start of sequence control is requested from the host PC **100**.

[0152] FIG. **14** is a block diagram illustrating a detailed configuration of the synchronization trigger distribution device **200**B. Since the configuration of the synchronization trigger distribution device **200**B is substantially the same as the configuration of the synchronization trigger distribution device **200**, redundant description will be omitted.

[0153] The configuration of the synchronization trigger distribution device **200**B is different from the configuration of the synchronization trigger distribution device **200** in the following points. That is, the table read control unit **220** is replaced with the table read control unit **220**B for which a sequence control start request received from the host PC **100** via the communication means **105** and the internal communication means **211** is corrected so as to be handled as a type of master trigger signal. Further, an execution time control unit **260** is newly added for subsequence execution time management, and the table read control unit **220**B receives an internal master trigger signal **261** output from the execution time control unit **260** instead of the master trigger signal **303**.

[0154] The execution time control unit **260** configured to measure an elapsed time from the beginning of the subsequence has a timer function including a time measurement counter and performs an operation similar to that of the delay correction unit. Specifically, when a timer enable signal **234** indicating whether the timer function is enabled or disabled indicates an enabled state and the timer start request signal **222** indicates a start request, a value corresponding to an execution time indicated by an execution time set value **235** is set as an initial value in the time measurement counter, and a decrement operation is performed based on a clock signal input (not shown). The execution time control unit **260**, which detects that the count value of the time measurement counter is zero, stops the decrement operation and outputs a pulse signal having a fixed assertion time width as the internal master trigger signal **261** to the table read control unit **220**B. The internal master trigger signal **261** is a signal inside the synchronization trigger distribution device **200**B, and there is no distribution trigger signal serving as a response signal to the internal master trigger signal **261** which is a request signal from the execution time control unit **260**.

[0155] Since the timer enable signal **234** and the execution time set value **235** can be output to the execution time control unit **260**, the control sequence table **230** is replaced with a control sequence table **230**B described later with reference to FIG. **15**. It is not necessary to change the response delay time table **240**.

[0156] FIG. **15** illustrates an example of settings in the control sequence table **230**B. The control sequence table includes a plurality of entries, and each entry identified by the entry number **2301** (corresponding to the count value of the trigger number counter **2201**) is associated with a specific subsequence. In addition to the distribution trigger signal output request fields **2302-1**, **2302-2**, **2302-3**, and **2302-4** for specifying whether the instrumentation devices **300**, **400**, **410**, and **420** should operate in the subsequence, that is, whether the distribution trigger signal output is necessary, and the auxiliary operation request field **2303** for specifying auxiliary operation contents to be processed by the table read control unit **220**B in parallel with the execution of the subsequence, one entry includes a master trigger signal output request field **2304** for specifying whether to request the execution of the subsequent subsequence after the execution time of the subsequence specified by the entry number **2301** is elapsed, that is, whether the output of the internal master trigger signal **261** is necessary, and an execution time set value field **2305** for specifying the execution time of the subsequence.

[0157] When a read request is received from the table read control unit **220**B via the read control signal **221**, setting information related to the subsequence is read from a specified entry position in the control sequence table **230**B, contents of the distribution trigger signal output request fields **2302-1**, **2302-2**, **2302-3**, and **2302-4** are output as the timer enable signals **231-1**, **231-2**, **231-3**, and **231-4**, respectively, contents of the auxiliary operation request field **2303** are output as the auxiliary operation control signal **232**, contents of the master trigger signal output request field **2304** are output as the timer enable signal **234**, and contents of the execution time set value field **2305** are output as the execution time set value **235**.

[0158] FIG. **16** is a time chart schematically illustrating a flow of processing when a control sequence of the quantum operation based on the settings in the control sequence table **230**B illustrated in FIG. **15** is executed in the quantum computing system **1**B according to the third embodiment. A main difference from the time chart in the first embodiment is that, by introducing the synchronization trigger distribution device **200**B which can generate the internal master trigger signal **261** replacing the master trigger signal **303** output by the instrumentation device **300** and to which an execution time management function is added, the synchronization trigger distribution device **200**B serves as a sequence control master device and centrally manages the overall sequence control following the start request. However, when there is processing to be executed as the preprocessing and post-processing subsequences, the processing is executed on the host PC **100** instead of the instrumentation device **300**.

[0159] When the initial setting (not shown) of the control sequence table **230**B and the response delay time table **240** is completed, the host PC **100** sets the trigger number counter **2201** in the table read control unit **220**B to zero via the communication means **105** and the internal communication means **211**.

[0160] Subsequently, as the last processing in the preprocessing subsequence, the host PC **100** requests the table read control unit **220**B to start sequence control via the communication means **105** and the internal communication means **211**.

[0161] The table read control unit **220**B, which receives the sequence control start request logically equivalent to the master trigger signal, shifts the operating state to the sequence execution state and performs increment in the trigger number counter. Further, the control sequence table **230**B is requested, via the read control signal **221**, to read setting information for the SEQ1 subsequence, and the execution time control unit **260** is requested, via the timer start request signal **222**, to start measuring the execution time of the SEQ1 subsequence.

[0162] When the timer enable signal **234** indicating "enabled or disabled" of the timer function in the SEQ1 subsequence output from the control sequence table **230**B indicates "enabled", the execution time control unit **260** receiving the request starts to measure the execution time, that is, decrement the time measurement counter, with the execution time set value **235** related to the SEQ1 subsequence as an initial value. The execution time control unit **260**, which detects that the count value of the time measurement counter is zero, requests the table read control unit **220**B to execute the subsequent SEQ2 subsequence via the internal master trigger signal **261**.

[0163] The table read control unit **220**B, which receives the request to execute the SEQ2 subsequence from the internal master trigger signal **261**, performs increment in the trigger number counter **2201**, requests the control sequence table **230**B to read the setting information for the SEQ2 subsequence, and requests the execution time control unit **260** to start measuring the execution time of the SEQ2 subsequence. According to the above procedure, sequence control start processing based on a request of the host PC **100** and switching of first execution target subsequences by the synchronization trigger distribution device **200**B alone following the start of the sequence control are completed. The subsequence switching is repeated up to the SEQ5 subsequence, and the host PC **100** is notified of the execution completion of the control sequence through the communication means **105** based on contents of the auxiliary operation control signal **232** in synchronization with switching to an SEQ6 subsequence.

[0164] In the third embodiment, although depending on the setting of a clock frequency for updating the time measurement counter provided in the execution time control unit **260**, the execution time measurement having time resolution several digits higher than the timer function of a normal operating system can be implemented. In addition, similarly to the second embodiment, since a master trigger signal output function is not required on the instrumentation device side, it is possible to achieve both cost reduction of the instrumentation device and highly accurate execution time management.

[0165] FIG. **17** is a refrigerator mounting diagram of a quantum computer system illustrating a mounting method using the quantum device **500** implemented by a quantum semiconductor, the instrumentation device **300** implemented by an analog chip, the host PC **100**, and a dilution refrigerator **1700** according to the first embodiment. Although not shown, the synchronization trigger distribution device **200** and the instrumentation devices **400**, **410**, and **420** are placed, for example, around the instrumentation device **300** in the dilution refrigerator **1700**.

[0166] In the dilution refrigerator **1700**, an air atmosphere outside the dilution refrigerator **1700** and a vacuum atmosphere inside the dilution refrigerator **1700** are separated by a housing **1701** and a room temperature plate **1702**. The degree of vacuum in the housing **1701** of the dilution refrigerator **1700** is controlled by discharging air through a vacuum tube VC using a pump device installed outside the dilution refrigerator **1700**. Temperature control in the dilution refrigerator **1700** is implemented by circulating diluted liquid helium in a pulse tube **1703** illustrated in FIG. **17**. FIG. **17** illustrates an example in which two pulse tubes **1703** are connected. The diluted liquid helium is obtained by liquefying two isotopes of helium, that is, ^{3}He and ^{4}He, respectively, and mixing ^{3}He phase with ^{4}He phase for dilution.

[0167] In the example of the dilution refrigerator **1700** in FIG. **17**, a plurality of metal (mainly oxygen-free copper) plates are provided and stored inside the housing **1701** of the dilution refrigerator **1700**. In this example, the metal plates used are 50K-PL (set to −223° C.), 4K-PL (set to −269° C.), PLA, PLB, and mKPL (set to approximately −273° C.). The temperature of the metal plates PLA and PLB is controlled in a range from 4K (−269° C.) to mK (about −273° C.). In the temperature control, a thermal equilibrium state is controlled and maintained by using a temperature control heater (not shown) mounted on each of the metal plates (50K-PL, 4K-PL, PLA, PLB, mKPL) and a temperature controller (not shown) installed outside the dilution refrigerator **1700** for controlling the amount of electric power supplied to the temperature control heater.

[0168] In the example in FIG. **17**, the diluted liquid helium is circulated from the pulse tube **1703** to a heat sink **1704**. Accordingly, the temperature of the metal plate 4K-PL and the metal plate mKPL to which the heat sink **1704** is connected is lowered to an extremely low temperature through the heat sink **1704** in which the diluted liquid helium circulates. Accordingly, the metal plate 4K-PL and the metal plate mKPL can be realized in an atmosphere of an extremely low temperature of 10 mK to 100 mK. The heat sink **1704** may be referred to as a first refrigerating tube, and the pulse tube **1703** may be referred to as a second refrigerating tube. The metal plate mKPL may be referred to as a first metal plate, and the metal plate 4K-PL may be referred to as a second metal plate.

[0169] The quantum device **500** is mounted on a first cooling plate FGNDPLT provided below the metal plate mKPL. The cooling plate FGNDPLT is thermally connected to the metal plate mKPL via cooling rods C**0** to C**3** (the even-numbered cooling rods C**0** and C**2** are not shown). That is, the heat sink **1704** is a refrigerating tube for cooling the cooling plate FGNDPLT, which is a metal body, using diluted liquid helium. The cooling plate FGNDPLT which is a metal body is thermally connected to the metal plate mKPL via the cooling rods C**0** to C**3**.

[0170] The reason why the quantum device **500** is not directly mounted on the metal plate mKPL but is installed below the metal plate mKPL in the example is to cause perform a quantum operation while applying a static magnetic field to the quantum device **500**. For the convenience of a space for providing a magnet **1705** for generating a static magnetic field at a lowermost layer of the dilution refrigerator **1700**, the quantum device **500** is provided as illustrated in FIG. **17** in the configuration example of the dilution refrigerator **1700** in the example. Control signals (**301**, **302**, **401**, **411**, and **521**) necessary for causing the quantum device **500** to perform a quantum operation are output from at least one of the instrumentation devices **300**, **400**, **410**, and **420**, and the control signals are electrically connected to the quantum device **500** via coaxial wires CXE and CXO.

[0171] As described above, since the quantum device **500**, the instrumentation device **300**, and the instrumentation devices **400**, **410**, and **420** are dispersedly arranged in the dilution refrigerator **1700** having a predetermined volume, the length of the wiring for connecting the quantum device **500**, the instrumentation device **300**, and the instrumentation devices **400**, **410**, and **420** is several tens of centimeters to several meters, and it is complicated to adjust the signal delay time.

[0172] While embodiments of the invention have been described above, these embodiments have been presented by way of example only, and are not intended to limit the scope of the invention. The novel embodiments can be carried out in various other forms, and various omissions, replacements, and modifications can be made without departing from the gist of the invention. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and gist of the invention. The components described in the embodiments may be implemented by dedicatedly designed hardware such as an application specific integrated circuit (ASIC), or may be implemented by programmable hardware such as a field-programmable gate array (FPGA). Further, at least some of the functions may be performed by software running on the host PC.

[0173] According to the above embodiments, since a practical quantum computer can be implemented, the energy consumption can be reduced and the carbon emission can be reduced, contributing to slowing of global warming and development of a sustainable society.

What is claimed is:

1. A quantum computing system comprising:

a quantum device configured to execute a quantum operation;

a plurality of instrumentation devices configured to transmit a control signal for executing the quantum operation to the quantum device; and

a synchronization trigger distribution device configured to transmit a second trigger signal for instructing transmission of the control signal to at least one of the plurality of instrumentation devices based on a first trigger signal indicating a timing, wherein

the synchronization trigger distribution device includes

a control sequence table for specifying a predetermined instrumentation device, to which the second trigger signal is to be output, among the plurality of instrumentation devices based on the first trigger signal, and

a delay time table for specifying a delay time corresponding to each of the plurality of instrumentation devices, and

outputs the second trigger signal to the instrumentation device specified based on the control sequence table, at a timing based on the delay time specified based on the delay time table.

2. The quantum computing system according to claim **1**, wherein
the synchronization trigger distribution device includes a table read control unit,
the table read control unit includes a trigger number counter configured to count the first trigger signal, and
the control sequence table specifies a predetermined instrumentation device, to which the second trigger signal is to be output, among the plurality of instrumentation devices based on a count number counted by the trigger number counter.

3. The quantum computing system according to claim **2**, wherein
the synchronization trigger distribution device receives a command for requesting the table read control unit to update a count value of the trigger number counter to a specified value.

4. The quantum computing system according to claim **2**, wherein
the synchronization trigger distribution device receives information specifying a processing state of the instrumentation device from at least one of the plurality of instrumentation devices.

5. The quantum computing system according to claim **1**, wherein
the synchronization trigger distribution device includes a delay correction unit corresponding to each of the plurality of instrumentation devices, and
when the instrumentation device corresponding to the delay correction unit is specified by the control sequence table, the delay correction unit corrects a timing based on the delay time of the instrumentation device corresponding to the delay correction unit, which is specified in the delay time table, and outputs the second trigger signal.

6. The quantum computing system according to claim **1**, wherein
the delay time is a delay time starting from the second trigger signal.

7. The quantum computing system according to claim **1**, wherein
the delay time includes at least one of a propagation delay of a transmission medium and a delay based on a communication protocol.

8. The quantum computing system according to claim **1**, comprising:
a cooling device, wherein
the quantum device is provided at a lowest-temperature portion in the cooling device, and
at least one of the plurality of instrumentation devices and the synchronization trigger distribution device is provided at a portion having a temperature different from the temperature of the portion where the quantum device is provided.

9. The quantum computing system according to claim **8**, wherein
the first trigger signal is generated by at least one of the plurality of instrumentation devices.

10. The quantum computing system according to claim **8**, wherein
the first trigger signal is specialized in a function related to the first trigger signal and is generated by a device that does not have a function related to the control signal and that is provided at a portion having a temperature different from that of a portion where the quantum device is provided.

11. The quantum computing system according to claim **8**, comprising:
a host device provided at a portion closest to room temperature in the system, wherein
the host device generates the first trigger signal.

12. The quantum computing system according to claim **8**, wherein
the first trigger signal is generated in the synchronization trigger distribution device.

13. An inter-device synchronization control method for quantum computing,
the method using
a quantum device configured to execute a quantum operation,
a plurality of instrumentation devices configured to transmit a control signal for executing the quantum operation to the quantum device, and
a synchronization trigger distribution device configured to transmit a second trigger signal for instructing transmission of the control signal to at least one of the plurality of instrumentation devices based on a first trigger signal indicating a timing,
the method comprising using the synchronization trigger distribution device to execute:
a first step of specifying a predetermined instrumentation device, to which the second trigger signal is to be input, among the plurality of instrumentation devices based on the first trigger signal;
a second step of specifying a delay time corresponding to each of the plurality of instrumentation devices; and
a third step of outputting the second trigger signal to the instrumentation device specified based on the first step, at a timing based on the delay time specified based on the second step.

14. The inter-device synchronization control method for quantum computing according to claim **13**, wherein
in the first step,
the first trigger signal is counted, and
a predetermined instrumentation device, to which the second trigger signal is to be output, is specified from among the plurality of instrumentation devices in accordance with a count value of the first trigger signal.

15. The inter-device synchronization control method for quantum computing according to claim **13**, wherein
the quantum device, the instrumentation device, and the synchronization trigger distribution device are provided in a cooling device, and
the first trigger signal is generated by at least one of the instrumentation devices, at least one of the synchronization trigger distribution device, at least one of other devices in the cooling device, and at least one of other devices outside the cooling device.

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